

ORDER FLOW SOFTWARE

Functional Specification v4.0

v4.0 adds Sections 9–23: Complete Indicator Calculation Formulas for all 26 indicators

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1. Complete Reference Registry

All sources consulted during the preparation of this specification — including the four PDF books provided by the user, the orderflows.com website, and all web sources independently researched — are catalogued below with their precise URLs.

1.1 Primary Source Documents (User-Supplied PDFs)

#	Title	Author	Year	Coverage
1	Trading Order Flow: Understanding & Profiting From Market Generated Information As It Occurs	Michael Valtos	2015	Footprint charts, Delta, COT, Imbalances, Stacked imbalance, Absorption, Trapped traders, Aggressive/passive participants, Initiative activity, VWAP, Iceberg orders, HFT, Candlestick + order flow, Chart types
2	ORDER FLOW: Trading Setups — Golden Ticket Edition	Dale (trader-dale.com)	2024	Volume clusters, HVN, Multiple nodes, Trades filter, Stacked imbalances, Unfinished business, Confirmation setups (big limit orders, absorption, aggressive orders, CVD divergence), Take profit/stop loss with order flow, Volume profile shapes, Volume profile setups
3	VOLUME PROFILE: The Insider's Guide to Trading — Golden Ticket Edition	Dale (trader-dale.com)	2024	Price action strategies, Volume profile POC/VAH/VAL, D/P/b/thin profile shapes, Volume accumulation setup, Trend setup, Rejection setup, Money management, Position sizing, Backtesting phases, Trading psychology
4	VWAP Book (vwap.com)	Dale (trader-dale.com)	2024	VWAP formula, Anchored VWAP (8 anchor types), VWAP strategies (reaction, rotation, trend), VWAP + Order Flow + Volume Profile confluences, Take profit placement, Stop loss placement, Trade trailing

1.2 Primary Website — orderflows.com (Newly Added)

 Fetched and fully read during preparation of v2.0. All pages listed below were directly accessed.

Page / Section	URL	Key Content Extracted
orderflows.com Home	https://www.orderflows.com/	Full product list, Pulse indicator summary, Turns indicator summary, Imbalance S&R summary, Single Prints summary, site navigation
Orderflows Pulse	https://www.orderflows.com/pulse.html	Full Pulse indicator specification: 7-variable composite (Order Flow, Delta, POC, Imbalances, Volume, Price Action, Swing Analysis) + Multi-Time Frame analysis upgrade, momentum jumps, order flow patterns
Orderflows Turns	https://www.orderflows.com/turns.html	Full Turns indicator specification: POC, bid/ask volume, volume, price action, delta, swing analysis. Works on footprint AND regular bar charts. Identifies turning points.
Imbalance Support & Resistance	https://www.orderflows.com/2015/08/22/imbalance-support-and-resistance/	Stacked imbalance zone indicator — draws buy/sell zones on chart, real-world analogy, 1–3 tick stops with 10x+ R/R potential
Top Heavy Big Seller / Bottom Heavy Big Buyer	https://www.orderflows.com/2015/08/22/big-buyer-seller/	Orderflows Ratio indicator — detects institutional big buyer at low or big seller at high. Acts as price floor/ceiling. Identifies institutional hedges and spreads.
Single Prints – Last Buyer / Last Seller	https://www.orderflows.com/2015/08/22/last-buyerseller/	Single print box indicator — marks last buyer at high / last seller at low. Signals end of move. Works on short and medium term charts.
Delta Surge (Free)	https://www.orderflows.com/deltasurge/index.html	Delta Surge indicator — analyzes delta to capitalize on real-time market information. Works on footprint AND normal bar charts. Free download. Delta trend analysis.
Orderflows Trader 7 Special	https://www.orderflows.com/special.html	Full list of 30+ automatable signals: Accumulation/Distribution, Open POC, Aligned POCs, Orderflows Gaps, Imbalance Reload, Volume Decline, POC Wave, Delta Tail, Resting Liquidity, Vertical Liquidity, Delta Breakout, Retail Suck, Price Action Divergence, Prominent POC, Orderflows Tails, Delta Divergence, Exhaustion Prints, Imbalance Reversals, Market Sweep, Market Weakness, Orderflows Sequencing, POC Slingshot, Zero Print, Engulfing Value Area, Inverse Imbalance, Stacked Imbalance, Thin Prints, Orderflows Ratio, Price Exhaustion,

		Price Defense, Multiple Imbalances
Orderflows Inner Circle Video Club	https://www.orderflows.com/videoclub/index.html	50+ webinar topics: Bar Delta Divergence (Webinar 18), 2nd Slot Imbalances (Webinar 19), Valtos U-Turn indicator (Webinar 23), Chart Types (Webinar 49), and many more.
Orderflows Toolbox 8.0	https://www.orderflows.com/toolbox/	Toolbox product — separate from NinjaTrader 8 basic Orderflows Toolkit. Lifetime license. No monthly fees.
Orderflows Software page	https://www.orderflows.com/software/	General software description, NinjaTrader integration details
Markets Traded	https://www.orderflows.com/markets-traded/	Works on any futures market with adequate volume: ES, NQ, YM, CL, GC, ZB, 6E, grains, energy

1.3 Web Sources Independently Researched (Not Supplied by User)

***i** All URLs below were found and read during research sessions — not provided by the user. These are additional reference sources used to build the specification.*

Source Name	URL	What It Contributed
NinjaTrader Official Blog — Order Flow / Footprint Charts (Oct 2025)	https://ninjatrade.com/futures/blogs/ninjatrade-order-flow/	Official definition of Footprint Chart vs Volumetric Bars; Order Flow+ suite description; Lee-Ready classification mention; industry-standard term confirmation
TradeThePool — Footprint Charts Explained (Oct 2025)	https://tradethepool.com/fundamental/footprint-charts-explained/	Visual definition of footprint profile; institutional interest zones; three-dimensional view of market activity
TradeThePool — Mastering Footprint Charts (Oct 2025)	https://tradethepool.com/fundamental/mastering-footprint-charts-trading/	Delta footprint definition; volume footprint definition; order flow footprint formal definition; unfinished auctions definition
LiteFinance — Order Flow Trading with Footprint Charts (Apr 2026)	https://www.litefinance.org/blog/for-beginners/trading-strategies/order-flow-trading-with-footprint-charts/	POC and Value Area definitions; passive vs aggressive order types; delta divergence as signal; delta trend analysis patterns
OptFutures — Footprint Charts Complete Guide	https://optimusfutures.com/blog/footprint-charts/	Three types of footprint charts (bid/ask, volume, delta); DOM explanation; stacked imbalances strategy; unfinished auctions strategy; image examples
Wikipedia — Order Flow Trading	https://en.wikipedia.org/wiki/Order_flow_trading	Buy/sell imbalance formal definition; spoofing/spoof orders definition; encyclopaedic reference for industry-

		standard terms
Triad City Beat — Footprint Charts for Beginners (Jan 2026)	https://triad-city-beat.com/footprint-charts-for-beginners-a-practical-guide-to-order-flow-trading/	Platform list (Sierra Chart, NinjaTrader, Bookmap, MT5); retail trader access to institutional tools; row-level price level description
TRADEPRO Academy — Footprint Chart Advantage	https://tradeproacademy.com/visualizing-market-order-flow-the-footprint-chart-advantage/	Range bar footprint example; WHO is in control framework; ES E-mini footprint chart example
GoCharting — Orderflows Trader User Guide (Apr 2026)	https://gocharting.com/docs/orderflow/orderflowtrader	Full Orderflows Trader platform documentation; stacked buying imbalance examples; delta candle divergence display; complete platform feature guide
ClusterDelta — Footprint for cTrader	https://clusterdelta.com/ctrader-footprint	Value Area Line Color settings; VAL/VAH line description; Profile BG inside/outside VA settings; cross-platform implementation reference
ClusterDelta — Footprint for MetaTrader 4/5	https://clusterdelta.com/footprint	Accumulation and distribution of market volumes; multi-platform compatibility reference; #Footprint indicator documentation
NordFX — Footprint Charts & Cumulative Delta (Nov 2025)	https://nordfx.com/en/useful-articles/footprint-charts-cumulative-delta-order-flow-trading	CVD divergence framework; combining footprint with classic price structure; demo account learning curve guidance; futures vs spot market correlation
Finowings — Order Flow Analysis: Footprint, Delta, Imbalance	https://www.finowings.com/Trading/order-flow-analysis-footprint-delta	Stacked imbalances support/resistance explanation; absorption definition; exhaustion definition; step-by-step order flow process for beginners
Wikipedia — NinjaTrader	https://en.wikipedia.org/wiki/NinjaTrader	TASC Readers Choice Award history (2021–2024); Kraken acquisition May 2025 for \$1.5B; platform history and brokerage info
Silver Creek Leadership — Mike Valtos Ultimate Bundle 30 (Feb 2026)	https://silvercreekleadership.academy/courses/orderflows-mike-valtos-ultimate-bundle-30/	Course overview; 30 modules; 2025 volatility spike updates; community Telegram group; ongoing update schedule
udcourse.com — Mike Valtos Bundle	https://udcourse.com/product/mike-valtos-bundle-30-orderflows-courses/	60–90 day screen time for tape reading mastery; Sierra Chart and NinjaTrader platform requirements; best markets (ES, NQ, CL, GC, ZB, BTC, ETH)
Orderflows Inner Circle Group	https://www.orderflows.com/innercircle/oldprice.html	Webinar 18 Bar Delta Divergence full definition; Webinar 19 2nd Slot Imbalances definition
orderflowstrader.com — Orderflows Trader 6	http://www.orderflowstrader.com/	8 new tools in v6: Thin Prints, Multiple Thin Prints, Delta/Volume Percentage, Extreme Delta/Volume, Extreme Delta,

		Small Min/Max Delta
Orderflows Trader 7 Special page	https://www.orderflows.com/special.html	Complete automatable signal list for v7; Markers Plus compatibility; third-party automation framework
Orderflows Toolbox page	https://www.orderflows.com/toolbox/	Toolbox vs Toolkit distinction; lifetime license model; no monthly fees policy
govinfo.gov — Citadel CEO Congressional Testimony (VWAP)	https://www.govinfo.gov/content/pkg/CHRG-117hhrg43966/pdf/CHRG-117hhrg43966.pdf	Official US Congressional testimony from Kenneth Griffin (Citadel CEO) confirming institutional VWAP program trading as virtually universal; VWAP slicing methodology
flowshifts.com — Michael Valtos background	http://www.flowshifts.com/	Confirmed 20-year institutional career: 8yr JP Morgan, 4yr Cargill, 3yr Commerzbank, 2yr EDF Man, 4yr Dean Witter Reynolds; aggressive vs passive framework explanation
TradingIQ — OrderFlow IQ (TradingView)	https://fr.tradingview.com/script/GX4WhZ8h-TradingIQ-OrderFlow-IQ	TradingView footprint implementation: POC, value area, imbalance detection, stacked imbalances, CVD levels across footprint

1.4 Reference Summary Count

Category	Count	Details
User-supplied PDF books	4	Valtos Trading Order Flow; Dale Order Flow Setups; Dale Volume Profile; Dale VWAP
orderflows.com pages (newly added)	12	Home, Pulse, Turns, Imbalance S&R, Big Buyer/Seller, Single Prints, Delta Surge, Trader 7, Inner Circle, Toolbox, Software, Markets
YouTube channel	1	@orderflows — channel metadata + video topic research
Independent web sources	23	NinjaTrader, TradeThePool ×2, LiteFinance, OptFutures, Wikipedia ×2, Triad City Beat, TRADEPRO, GoCharting, ClusterDelta ×2, NordfX, Finowings, Silver Creek, udcourse, orderflowstrader, govinfo.gov, flowshifts, TradingIQ, Inner Circle
TOTAL	40	All sources catalogued above

2. orderflows.com Indicators — Full Specification

The following indicators are Michael Valtos's proprietary products sold at orderflows.com. They are directly relevant to the software we are building — they define the signals and features that the commercial product must detect and surface. All specifications below are derived from the orderflows.com website pages fetched directly.

2.1 Orderflows Pulse

Source: <https://www.orderflows.com/pulse.html> — fetched directly

Description: The Orderflows Pulse is Valtos's most comprehensive composite indicator. It analyzes 7 order flow variables simultaneously and generates trading signals on both footprint charts AND regular candlestick charts. It received a major upgrade with Multi-Time Frame analysis, allowing higher-timeframe order flow signals to be overlaid on lower-timeframe charts.

2.1.1 The 7 Analysis Variables

Variable	What It Analyzes	Signal Contribution
Order Flow	What is currently trading and HOW it is trading in real-time	Identifies order flow patterns and execution aggression
Delta	Net difference between aggressive buyers and aggressive sellers	Shows which side is dominating the current bar
Point of Control (POC)	Price level with highest volume in the bar — only IMPORTANT POCs are used	Identifies institutional consensus price; POC slingshot and defense
Imbalances	Price levels where aggressive traders outnumber others by large percentages	Detects supply/demand imbalance; generates buy/sell signals
Volume	Sufficient volume threshold required for accurate assessment	Filters noise bars; only signals on bars with meaningful volume
Price Action	How price is reacting to the order flow context	Confirms or negates the order flow signal with price behavior
Swing Analysis	Reference to recent significant swing highs and swing lows	Provides context for current order flow relative to key market structure

2.1.2 Multi-Time Frame Analysis (Major Upgrade)

- Allows higher-timeframe order flow signals to be displayed on the current chart without switching timeframes
- Example: trading a 30-second ES chart — add signals from the 3-minute ES chart on top
- Requires the same 7-variable analysis to pass on BOTH timeframes for highest-conviction signal
- Works on any chart type: minute-based, volume-based, tick-based, range-based

2.1.3 Pulse Signal Types

Signal Type	Definition	Trading Application
Momentum Jump	Market momentum surges at a specific price — 'market sweet spot'. Optimal entry before and during trends.	Enter in direction of momentum jump; high-probability trend entry
Order Flow Pattern	Real-time analysis of how orders are executing — identifies immediate market dynamics	Confirms short-term directional bias based on live order activity
Delta Imbalance	Net aggressive buying vs selling dominance detected at the bar level	Identifies which side controls the bar; filters setups by direction
Volume Concentration	POC identification — highest volume price level in the bar	Finds institutional activity center; used for entry/stop placement
Price Imbalance	Aggressive trader concentration significantly outnumbers opposing side at a price level	Identifies likely support/resistance from the order flow data
Price Action Reaction	How price responds to the detected order flow conditions	Confirms or invalidates signal; filters false signals
Swing Reference	Context from recent swing highs and lows	Gives market structure context to order flow signal

2.1.4 Platform & Technical Details

- Runs on NinjaTrader 8 as an add-on indicator
- Install time: approximately 25 seconds
- Compatible with The Indicator Store's Markers Plus (third-party automation)
- Works on footprint charts (Orderflows Trader) AND regular bar/candlestick charts
- Works on minute, volume, tick, and range-based chart types with same settings
- Price: \$225 (one-time, 40% discount from \$397)

2.2 Orderflows Turns

Source: <https://www.orderflows.com/turns.html> — fetched directly

Description: The Orderflows Turns is designed specifically to identify market TURNING POINTS using a combination of 6 order flow and market analysis variables. It is simpler than the Pulse and more focused on reversal detection. Works on both footprint charts and normal bar/candlestick charts.

2.2.1 The 6 Analysis Variables

Variable	What It Analyzes
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Point of Control	Most-traded price within the bar — where institutional activity was concentrated
Volume (Bid & Ask)	Volume traded on bid side vs ask side — detects aggressive order dominance
Volume (Total)	Sufficient total volume threshold to validate signal quality
Price Action	How price is behaving relative to the order flow — confirms or denies the potential turn
Delta	Net buyer vs seller aggression — is the turn confirmed by delta reversal?
Swing Analysis	Recent swing high/low reference points — is price at a significant structural level?

2.2.2 Turns Signal Behavior

- Identifies areas of the chart being defended or supported by institutional activity
- Like having Valtos 'looking over your shoulder' pointing out defended/supported areas
- NOT a simple price pattern — combines multiple order flow data points
- Can be overlaid on top of existing footprint charts OR existing candlestick charts
- Designed for short-term day traders: recommended 4–10 range charts, timeframes under 5 minutes to 30 seconds
- Works on: ES, CL, YM, NQ, 6E, ZB, FGBL and all major liquid futures markets
- Not recommended for thinly-traded markets (Lean Hogs, Live Cattle)
- Platform: NinjaTrader 8 version 15+. Price: \$500 one-time.

2.3 Orderflows Ratio — Top Heavy Big Seller / Bottom Heavy Big Buyer

Source: <https://www.orderflows.com/2015/08/22/big-buyer-seller/> — fetched directly

Description: The Orderflows Ratio indicator detects the presence of institutional players so large that their activity creates short-term price floors and ceilings. Based on Valtos's personal experience placing 1000+ lot institutional orders.

2.3.1 Signal Logic

- Bottom Heavy Big Buyer: detected at a LOW — a large institutional buyer absorbs all selling at a price level, acting as a price FLOOR
- Top Heavy Big Seller: detected at a HIGH — a large institutional seller satisfies all buyers at a price level, acting as a price CEILING
- Based on bid/ask RATIO across the bar — the ask/bid ratio exceeds a configured threshold indicating one side is massively dominant
- May represent commercial hedges or spread trades — Valtos's specific use case from his institutional trading career

2.3.2 Trading Application

- When Bottom Heavy Big Buyer appears at a low: buy the retest; stop below the detected level
- When Top Heavy Big Seller appears at a high: sell the retest; stop above the detected level
- 1–3 tick stops with profit potential 10x+ the risk (Valtos's stated specification)
- Works on the same markets as other Orderflows indicators — any liquid futures

2.4 Single Prints — Last Buyer / Last Seller

Source: <https://www.orderflows.com/2015/08/22/last-buyerseller/> — fetched directly

Description: The Single Print Box indicator identifies when the LAST buyer has bought at a high or the LAST seller has sold at a low. This marks the potential END of a trend leg — nobody wants to transact at that extreme anymore.

2.4.1 Signal Logic

- Last Buyer at High: only 1–2 trades executed at the high — demand has dried up at that price. Price will begin to fall.
- Last Seller at Low: only 1–2 trades at the low — supply has dried up at that price. Price will begin to rise.
- The single print (low/zero volume at extreme) is the visual indicator on the footprint chart
- Works as both scalp (short-term) and swing (medium-term) trade signal
- The indicator draws a box around the single print for easy visual identification

***i** This is the orderflows.com implementation of what Dale calls 'Unfinished Business' / 'Failed Auction'. Same concept, different naming convention.*

2.5 Delta Surge (Free Indicator)

Source: <https://www.orderflows.com/deltasurge/index.html> — fetched directly

Description: Delta Surge is a FREE indicator from Valtos that analyzes order flow delta and works on BOTH footprint charts AND normal bar charts. Its purpose is to help traders who are not yet using footprint charts to access delta analysis.

2.5.1 Signal Logic

- Analyzes delta trend — in a healthy uptrend, delta builds on itself as new buyers are drawn in
- Detects surges in delta that signal momentum acceleration in the prevailing direction
- Works independently of footprint chart — reads delta data from the instrument feed
- Allows delta analysis on standard candlestick charts without requiring Orderflows Trader
- Download: free at <https://www.orderflows.com/deltasurge/index.html>

- Use case: first step for traders transitioning from traditional charts to order flow analysis

2.6 Orderflows Indicators — Complete Feature Matrix for Implementation

The following table maps all orderflows.com indicators to required software features and signals the C++ engine must support:

Indicator	Required Engine Signals	Output to API	Priority
Pulse	Delta, POC, Imbalances, Volume threshold, Price action context, Swing H/L, Multi-TF analysis	REST: /v1/signals/pulse WS: signals stream	P1 — v1.0
Turns	POC, Bid/Ask ratio, Volume, Price action, Delta, Swing H/L, Turning point detection	REST: /v1/signals/turns WS: signals stream	P1 — v1.0
Orderflows Ratio (Big Buyer/Seller)	Bar-level bid/ask ratio exceeding threshold at high/low; volume confirmation	REST: /v1/signals/ratio WS: signals stream	P1 — v1.0
Single Prints (Last Buyer/Seller)	Zero or near-zero volume at bar extreme; single print box detection	REST: /v1/signals/single_prints WS: signals stream	P1 — v1.0
Delta Surge	Delta trend acceleration; delta surge threshold detection; works on all chart types	REST: /v1/signals/delta_surge WS: delta stream	P1 — v1.0
Stacked Imbalance Zones	3+ consecutive imbalances; zone drawing; persistence across bars	REST: /v1/zones WS: signals stream	P1 — v1.0
All Trader 7 signals (30+)	Full signal set per Trader 7 spec (Accumulation, POC Wave, Delta Tail, etc.)	REST: /v1/signals WS: signals stream	P2 — v1.5

3. Volume Profile as a Standalone Trading Instrument

Volume Profile is not merely a supporting indicator — it is a complete, standalone analytical instrument with its own data model, display specification, trading setups, and API surface. This section specifies Volume Profile as a first-class instrument in the software.

3.1 What Makes Volume Profile a Standalone Instrument

- Volume Profile provides the STRUCTURAL CONTEXT within which all order flow signals are interpreted
- Without Volume Profile, order flow signals have no reference frame — a stacked imbalance in the middle of a value area means something different from one at the POC
- Institutional traders rely on Volume Profile (Market Profile) to identify fair value, value area, and initiative vs responsive activity levels
- Dale explicitly states: 'I don't look for candlestick formations... I perceive the price movement as a continuous flow' — Volume Profile IS that continuous flow framework
- Volume Profile can be traded completely independently of order flow footprint charts — Dale's original approach before adding order flow

3.2 Volume Profile Data Model

Data Element	Type	Calculation	Significance
POC — Point of Control	Price level	Price with highest total volume in the profile period	Most-traded price = institutional consensus of fair value. Strongest S&R level.
VAH — Value Area High	Price level	Upper boundary of 70% volume range above POC	Upper edge of institutional comfort zone. Strong resistance.
VAL — Value Area Low	Price level	Lower boundary of 70% volume range below POC	Lower edge of institutional comfort zone. Strong support.
Value Area	Price range [VAL, VAH]	70% of all traded volume falls within this range	Zone of price acceptance. Two-sided trade. Most likely for price to return here.
HVN — High Volume Node	Price cluster	Volume significantly > average for adjacent levels	Institutional accumulation/distribution zone. Strong S&R. Price spends time here.
LVN — Low Volume Node	Price gap	Volume significantly < average for adjacent levels	Price acceptance gap. Price moves quickly through these levels.
Volume Distribution	Histogram	Volume at each price level plotted horizontally	Full picture of institutional activity across all prices in

			the period
Bid/Ask Split Profile	Dual histogram	Ask volume (buy) vs Bid volume (sell) per price level	Shows directional bias within the profile — which side is more aggressive at each price

3.3 Volume Profile Types (All Must Be Implemented)

Profile Type	Period Definition	Reset Behavior	Primary Use Case	Timeframe
Session Profile	Single RTH session (e.g., 9:30am–4:00pm ET)	Resets each session open	Intraday trading — identifies today's fair value and value area	1min–30min
Daily Profile	One calendar day (including extended hours)	Resets at midnight	Daily structure analysis; previous day POC as reference level	5min–1hr
Weekly Profile	Monday open to Friday close	Resets each Monday	Swing trading context; weekly fair value benchmark	30min–4hr
Monthly Profile	Calendar month	Resets each month	Institutional positioning; long-term supply/demand map	Daily
Yearly Profile	January 1 to December 31	Resets each January 1	Annual fair value; VWAP alignment analysis	Daily–Weekly
Flexible / Anchored Profile	User-defined start date to current bar	Manual anchor — does not reset	Custom analysis: earnings events, regime changes, macro news	Any
Composite Profile	Merged across multiple sessions	Does not reset — accumulates	Long-term institutional level mapping; multi-day absorption zones	Any
Real-Time Profile	Current session building in real time	Continuous update each tick	Live trading — watch POC migration, value area expansion	Any

3.4 Volume Profile Shape Classification Engine

The engine must classify the current profile shape automatically and expose the classification via API. Shape determines the expected market behavior.

Shape Name	Visual Pattern	Mathematical Definition	Market Interpretation	Expected Next Move
D-Profile (Bell	Wide symmetric	Standard deviation of	Balanced market. Two-	Continue

Curve)	distribution centered on POC	volume distribution is low; POC is near the median price; tail volume < 15%	sided trade. Neither buyers nor sellers dominating.	rotation within value area until imbalance breaks out
P-Profile (Buy Imbalance)	Narrow base, wide top; POC is near upper 30% of range	Volume skewed to upper prices; lower prices have thin volume tail	Strong buying occurred near highs; accumulation at lows early, distribution later.	Continuation UP if holding above POC; short on failure below POC
b-Profile (Sell Imbalance)	Wide base, narrow top; POC is near lower 30% of range	Volume skewed to lower prices; upper prices have thin volume tail	Strong selling occurred near lows; distribution at highs early, accumulation at lows.	Continuation DOWN if holding below POC; long on failure above POC
Thin Profile	Narrow, elongated; minimal volume at most levels	Standard deviation of volume distribution is high; no clear POC concentration	Trending/initiative market. Price moved quickly. Little consensus.	Fast reversion to value if move is responsive; fast continuation if initiative

3.5 Volume Profile Trading Setups (Standalone)

Source: Dale — Volume Profile Insider's Guide; Dale — Order Flow Trading Setups ch. Volume Profile

Setup VP-1: Volume Accumulation Setup (D-Profile Breakout)

- Identify a D-profile (balanced, bell-curve) area — price has been rotating inside value area
- Wait for price to break OUT of the value area with conviction (bar closing clearly above VAH or below VAL)
- Entry trigger: first bar closing outside value area with delta 3x or more above average
- Stop: back inside the value area (VAH becomes support for long; VAL becomes resistance for short)
- Target: next significant HVN, POC, or VWAP level beyond the breakout
- Logic: the D-profile represents institutional accumulation — breakout signals their distribution/initiative

Setup VP-2: Trend Setup (P/b-Profile Continuation)

- Identify a P-profile (bullish) or b-profile (bearish) — shows directional institutional intent
- P-profile: wait for pullback to POC or VAH from above; look for long entry
- b-profile: wait for rally to POC or VAL from below; look for short entry

- Entry trigger: delta confirmation at the pullback level; delta should be in trend direction
- Stop: clearly on wrong side of POC (below for longs in P-profile; above for shorts in b-profile)
- Target: previous swing high/low; next week's VWAP; next profile POC
- Logic: POC is institutional consensus — they will defend it in trend direction

Setup VP-3: Rejection Setup (Thin Profile Reversal)

- Identify a thin profile area — price moved quickly through with little volume = no institutional acceptance
- Price reaches the thin profile extreme and shows aggressive rejection delta
- Entry trigger: rejection bar with strong opposing delta; CVD divergence at thin profile extreme
- Stop: beyond the thin profile extreme (1–2 ticks past the zero-volume zone)
- Target: back to the origin of the thin profile (nearest HVN or previous session POC)
- Logic: institutions did not transact here — no support/resistance. Price snaps back to value.

Setup VP-4: Value Area Re-Entry (Responsive Trade)

- Price breaks BELOW VAL or ABOVE VAH but with weak or negative delta — responsive move
- Price rapidly reverses back inside the value area
- Entry: on return into value area at VAL/VAH with confirming order flow
- Stop: beyond the recent extreme (outside the value area)
- Target: POC (value area center) — responsive trades target a return to fair value
- Logic: institutions did not support the breakout — they faded it with limit orders

Setup VP-5: POC as Dynamic Support/Resistance

- The POC of the previous session/day/week acts as a reference level for the next session
- Previous Session POC (VPOC): if current price is above, acts as support; if below, acts as resistance
- Strong signal when current session volume builds around the previous VPOC — institutional memory
- Entry: test of previous VPOC with order flow confirmation (absorption or buying imbalance at support VPOC)
- Stop: break below previous VPOC with strong sell delta
- Target: current session VAH or VAL

Setup VP-6: Volume Profile + Order Flow Confluence (Highest Conviction)

- Find a Volume Profile key level (POC, VAH, VAL, HVN) — this is the structural setup

- Wait for price to arrive at the level
- Require order flow confirmation: stacked imbalance OR absorption OR delta divergence at the level
- Also require VWAP alignment: price at VP level that also coincides with Daily or Weekly VWAP
- Triple confluence (VP + Order Flow + VWAP) = highest conviction trade signal in the system
- Entry: at the confluence level; stop 1–3 ticks outside; target 2:1 or better R/R to next structure

***i** This is the core methodology that both Valtos and Dale describe as their preferred approach. Three-way confluence is the gold standard.*

3.6 Volume Profile API Surface

Endpoint	Method	Parameters	Response
/v1/profile/{instrument_id}	GET	period_type, start_date, end_date, session_filter	{ poc, vah, val, value_area_pct, hvns[], lvns[], shape, distribution[] }
/v1/profile/{instrument_id}/live	GET	period_type	Real-time current session profile building
/v1/profile/{instrument_id}/composite	GET	from_date, to_date	Composite profile across multiple sessions
/v1/profile/{instrument_id}/anchored	GET	anchor_timestamp	Flexible anchored profile from any point to current bar
WS: profile/{instrument}	WebSoc ket	Subscribe to profile type	Live updates: { poc, vah, val, shape, updated_at } on each bar close

3.7 Volume Profile Display Specification

Display Element	Default Color	Configurable	Notes
Volume histogram bars	Blue gradient (ask-heavy) / Red gradient (bid-heavy)	Yes	Darker = higher volume. Optional bid/ask split coloring.
POC line	Yellow horizontal line across chart	Yes	Most prominent line on the profile
VAH line	Green dashed horizontal line	Yes	Upper value area boundary
VAL line	Red dashed horizontal line	Yes	Lower value area boundary
Value area fill	Semi-transparent blue	Yes	Shaded region between VAL and VAH

HVN markers	Bold/wider volume bar	Yes	HVN automatically detected and highlighted
LVN markers	Thin/barely visible gap	Yes	LVN gaps automatically highlighted
Shape label	Text overlay: D / P / b / Thin	Yes	Auto-classified shape label displayed on chart
Profile type label	Text: Session / Daily / Weekly / etc.	Yes	Shows which profile type is displayed

3.8 Volume Profile Configuration Parameters

Parameter	Default	Range	Description
value_area_percent	70	60–90%	Percentage of volume defining value area (academic standard is 70%)
profile_type	SESSION	SESSION/ DAILY/ WEEKLY/ MONTHLY/ YEARLY/ FLEXIBLE/ COMPOSITE	Which profile type to display
show_multiple_profiles	false	true/false	Show multiple profiles simultaneously (e.g., session + weekly)
hvn_threshold_multiplier	1.5	1.1–3.0	Volume must exceed average × multiplier to qualify as HVN
lvn_threshold_multiplier	0.5	0.1–0.9	Volume must be less than average × multiplier to qualify as LVN
bid_ask_split	false	true/false	Show bid and ask volume separately in profile histogram
profile_width_pct	20	5–50%	Profile histogram width as percentage of chart width
auto_classify_shape	true	true/false	Auto-detect and label D/P/b/Thin profile shape
show_previous_poc	true	true/false	Display previous session/period POC as reference line
composite_lookback_days	20	5–252	Number of days to include in composite profile
real_time_update_ms	100	100–5000ms	How often live profile recalculates during session

4. Updated Complete Feature Matrix

Full list of all features across Order Flow, Volume Profile, VWAP, and orderflows.com Indicators that the software engine must implement.

Feature Category	Feature	Source	Version
Footprint	Bid/Ask volume per price level	Valtos book; Dale book; NinjaTrader	v1.0
Footprint	Cell color (green/red) by ask vs bid dominance	Valtos; Dale	v1.0
Footprint	Volume heat map (darker = higher)	Dale	v1.0
Footprint	Diagonal imbalance comparison	Valtos; Dale	v1.0
Delta	Bar delta (ask_vol - bid_vol)	Valtos; Dale; GoCharting	v1.0
Delta	Max delta / Min delta within bar	Valtos	v1.0
Delta	Cumulative Delta (CVD) — session total	Dale; GoCharting	v1.0
Delta	Delta/Volume percentage	Orderflows Trader 6	v1.0
Delta	Extreme delta detection	Orderflows Trader 6	v1.0
Delta	Small Min/Max delta detection	Orderflows Trader 6	v1.0
Delta	Delta Surge indicator	orderflows.com/deltasurge	v1.0
Delta Divergence	Standard: negative delta at new high / positive at new low	Valtos; Dale	v1.0
Delta Divergence	Bar Delta Divergence: bearish bar + positive delta	Valtos Webinar 18	v1.0
Delta Divergence	CVD Divergence: price vs cumulative delta	Dale	v1.0
Imbalance	Single buying/selling imbalance (2.5x/3x/4x)	Valtos; Dale; OptFutures	v1.0
Imbalance	Stacked imbalance (3+) zone with zone drawing	Valtos; Dale; orderflows.com	v1.0
Imbalance	2nd Slot imbalance detection	Valtos Webinar 19	v1.0
Imbalance	Imbalance reversal	Orderflows Trader 7	v1.0
Imbalance	Multiple imbalances cluster	Orderflows Trader 6/7	v1.0
Imbalance	Inverse imbalance	Orderflows Trader 6	v1.0
Imbalance	Imbalance reload	Orderflows Trader 7	v1.0
COT	Intrabar COT: highest volume price level in bar with box display	Valtos book	v1.0

Absorption	High volume + low net delta at price level	Valtos; Dale	v1.0
Trapped Traders	Imbalance at bar extreme in opposite direction to bar close	Valtos book	v1.0
Initiative/ Responsive	Price vs value area + delta at breakout	Valtos book	v1.0
Single Prints	Last buyer at high / last seller at low detection	orderflows.com/last-buyerseller	v1.0
Orderflows Ratio	Top heavy big seller / bottom heavy big buyer	orderflows.com/big-buyer-seller	v1.0
Orderflows Pulse	7-variable composite signal + multi-TF analysis	orderflows.com/pulse.html	v1.0
Orderflows Turns	6-variable turning point detector	orderflows.com/turns.html	v1.0
Volume Profile	Session profile with POC/VAH/VAL/HVN/LVN	Dale VP book; orderflows.com	v1.0
Volume Profile	Daily/Weekly/Monthly/Yearly profiles	Dale VP book	v1.0
Volume Profile	Flexible / Anchored profile	Dale VP book	v1.0
Volume Profile	Composite profile (multi-session)	Dale VP book	v1.0
Volume Profile	D/P/b/Thin shape auto-classification	Dale VP book	v1.0
Volume Profile	Prominent POC + Aligned POCs + Open POC	Orderflows Trader 7	v1.0
Volume Profile	POC Wave + POC Slingshot	Orderflows Trader 7	v1.0
Volume Profile	Bid/Ask split profile histogram	Dale book	v1.0
Volume Profile	VP Setup 1–6 signal detection	Dale; this spec §3.5	v1.0
VWAP	Daily/Weekly/Yearly VWAP	Dale VWAP book	v1.0
VWAP	Anchored VWAP (8 anchor types)	Dale VWAP book	v1.0
VWAP	Standard deviation bands (1σ , 2σ)	Dale VWAP book	v1.0
VWAP	VWAP reaction/rotation/trend setups	Dale VWAP book	v1.0
Trader 7 Signals	All 30+ automatable signals (Accumulation thru Multiple Imbalances)	Orderflows Trader 7 spec	v1.5
Market Structure	Session high/low, previous day high/low	Valtos book	v1.0
Market Structure	Unfinished Business / Failed Auction detector	Dale book; orderflows.com	v1.0
Chart Types	TIME / VOLUME / RANGE bar construction	Valtos book	v1.0

5. Implementation Notes — v2.0 Additions

5.1 Volume Profile as Primary Data Layer

Volume Profile data must be calculated and available BEFORE order flow signals are evaluated. The signal engine depends on VP context for signal validity:

- Stacked imbalance at VAH is stronger than stacked imbalance in middle of value area
- Delta divergence AT the POC is a high-conviction signal; delta divergence away from POC is lower conviction
- Absorption at the POC has different meaning than absorption at an HVN vs an LVN
- All 6 VP trading setups must be cross-referenced with order flow signals before alerting the user

5.2 orderflows.com Indicator Compatibility

 *The Pulse and Turns indicators are sold as commercial NinjaTrader products by Valtos. We are building our OWN implementation of these signal types, not reverse-engineering his protected code. Our signals follow the same analytical logic as documented on orderflows.com.*

- Pulse implementation: combine all 7 variables in a weighted scoring engine; threshold score triggers signal
- Turns implementation: 6-variable analysis with turning point pattern recognition
- Ratio implementation: bar-level bid/ask ratio at price extremes with configurable threshold
- Single Prints implementation: zero-print / last-print detection at bar highs/lows
- Delta Surge implementation: delta acceleration detection using rolling delta window

5.3 Multi-Instrument Volume Profile

Per Valtos's teaching: the same Volume Profile concepts apply across ALL futures markets. The software must support configuring profiles for:

- Index futures: ES (S&P 500), NQ (Nasdaq 100), YM (Dow Jones), RTY (Russell 2000)
- Energy: CL (Crude Oil), NG (Natural Gas), RB (RBOB Gasoline)
- Metals: GC (Gold), SI (Silver), HG (Copper)
- Fixed income: ZB (30-yr T-Bond), ZN (10-yr T-Note), ZF (5-yr T-Note)
- FX futures: 6E (Euro), 6B (British Pound), 6J (Japanese Yen), 6A (Australian Dollar)
- Agricultural: ZC (Corn), ZS (Soybeans), ZW (Wheat)
- Crypto futures: BTC, ETH (per udcourse.com reference)

 *Volume Profile parameters (value_area_percent, HVN/LVN thresholds) may need instrument-specific tuning — ES has different volume distribution characteristics than CL or ZB.*

5.4 Revised Implementation Priority

Phase	Timeline	Key Additions from v2.0
Phase 0: Interfaces	Week 1	Add Volume Profile module headers; add orderflows.com indicator interface headers
Phase 1: Core Engine	Week 2–4	Tick ingestion, bar construction, delta engine, basic footprint — unchanged
Phase 2: Volume Profile Engine	Week 4–6	MOVED UP: Volume Profile is now Phase 2 (was Phase 3) — required as foundation for signal context
Phase 3: Order Flow Analytics	Week 6–8	Full imbalance suite, absorption, COT, trapped traders — now uses VP context
Phase 4: orderflows.com Signals	Week 8–10	Pulse, Turns, Ratio, Single Prints, Delta Surge implementations
Phase 5: VWAP + Advanced Signals	Week 10–12	VWAP engine; all VWAP strategies; Trader 7 signals subset
Phase 6: API Layer	Week 12–14	REST + WebSocket + VP API surface + orderflows.com signal endpoints
Phase 7: NinjaTrader + Testing	Week 14–18	NinjaTrader adapter; full test suite; historical replay
Phase 8: Security + Launch	Week 18–22	License engine; obfuscation; Paddle integration; NinjaTrader v1.0 public release

6. Additional Glossary Entries (v2.0)

Term	Definition	Source
Orderflows Pulse	Composite order flow indicator analyzing 7 variables (Order Flow, Delta, POC, Imbalances, Volume, Price Action, Swing Analysis) with multi-timeframe capability. Generates momentum jump, delta imbalance, and price reaction signals.	orderflows.com/pulse.html
Orderflows Turns	Order flow turning point detector analyzing 6 variables (POC, Bid/Ask volume, Volume, Price Action, Delta, Swing Analysis). Identifies defended/supported price areas for reversal entries.	orderflows.com/turns.html
Orderflows Ratio	Indicator detecting institutional-scale buyers (Bottom Heavy) at lows or sellers (Top Heavy) at highs — based on bid/ask ratio at price extremes.	orderflows.com/big-buyer-seller
Single Print / Last Buyer / Last Seller	Zero or near-zero volume at a bar extreme — signals the last trade at that price. Marks potential end of move. Orderflows implementation of Unfinished Business concept.	orderflows.com/last-buyerseller
Delta Surge	Free indicator detecting surges in delta trend. Works on both footprint and regular bar charts. Signals momentum acceleration.	orderflows.com/deltasurge
Bottom Heavy	Bar pattern where the majority of volume and bid/ask activity concentrates at the LOWER price levels — indicates strong passive or aggressive buying at lows.	orderflows.com/big-buyer-seller
Top Heavy	Bar pattern where the majority of volume and bid/ask activity concentrates at the UPPER price levels — indicates strong passive or aggressive selling at highs.	orderflows.com/big-buyer-seller
Momentum Jump	Pulse signal: sudden acceleration in order flow momentum at a specific price — identified as optimal entry point.	orderflows.com/pulse.html
Standalone Volume Profile	Volume Profile used as the primary trading instrument rather than a supporting indicator. Full analytical framework including profile types, shape classification, and 6 trading setups.	This spec §3; Dale VP book
D-Profile	Bell-curve shaped volume profile — balanced, rotational market with two-sided trade facilitation.	Dale Volume Profile book
P-Profile	Volume profile with concentration at upper prices (narrow base, wide top) — bullish accumulation/distribution pattern.	Dale Volume Profile book
b-Profile	Volume profile with concentration at lower prices (wide base, narrow top) — bearish distribution/accumulation pattern.	Dale Volume Profile book

Triple Confluence	Highest-conviction setup: Volume Profile key level + Order Flow signal (imbalance/absorption) + VWAP level all aligning at the same price.	Dale VWAP book; this spec §3.5
Value Area Re-Entry	Trade setup: price breaks out of the value area on weak/responsive delta and rapidly returns inside — fade the breakout.	This spec §3.5 Setup VP-4
Previous Session POC	The POC from the prior session that carries forward as a reference level for the current session — institutional memory level.	This spec §3.5 Setup VP-5

7. Bar & Candlestick Time Configuration

Every bar (candlestick) in the Order Flow Engine is fully time-configurable. 'Bar' and 'candlestick' refer to the same unit of market data — the difference is only visual. A bar shows OHLC as lines; a candlestick shows the same data as a coloured body with wicks. The underlying data is identical and all order flow metrics (delta, imbalance, footprint, COT) are computed identically regardless of which visual style is selected.

7.1 Bar Construction Methods

The engine supports three fundamentally different methods of constructing bars. The choice determines WHEN a new bar opens and closes:

Method	New Bar Opens When...	New Bar Closes When...	Valtos Preference	Best For
TIME	A clock boundary is crossed (e.g. every 5 minutes on the minute)	The next clock boundary arrives	3rd choice	Alignment with economic events; most trader-familiar; easy to compare sessions
RANGE	The previous bar's price range is complete (e.g. 4 ticks of movement)	Price moves outside the configured tick range from bar open	1st choice (primary)	Volatility-adaptive; new bars only form when price MOVES; clean order flow signal isolation
VOLUME	The previous bar accumulated the configured contract count	The running volume total reaches the configured threshold	2nd choice	Equal-volume bars; institutional activity analysis; size-normalised footprints

***i** Valtos explicitly states in the source book: 'Given a choice between range based charts, volume based charts and time based charts I would choose range based charts followed by volume charts followed by time based charts.' However all three methods are fully configurable by the user.*

7.2 TIME-Based Bar Sizes — Complete Configuration Table

For TIME-based bars, the user selects a time duration. Every valid duration the engine supports is listed below across three tiers: sub-minute (seconds), minute-based, and hour/session-based.

7.2.1 Sub-Minute (Seconds) — Tier 1

Bar Size	Config Value	Bars Per RTH Session (6.5hr)	Typical Use Case	Order Flow Notes
5 seconds	5s	4,680	High-frequency scalping; HFT pattern analysis	Extremely noisy footprint. Only useful for liquid markets (ES, NQ) with 50,000+ contracts/session. Delta very small per bar.
10 seconds	10s	2,340	Scalping; short-term momentum; tape reading	Still noisy but manageable. Begin to see meaningful imbalances at high-volume moments.
15 seconds	15s	1,560	Scalping; fast intraday momentum	Better imbalance signal quality. Pulse and Turns signals become more reliable than 5s/10s.
20 seconds	20s	1,170	Scalping; short-term delta divergence	Less common but valid. Use when 15s is too noisy and 30s is too slow.
30 seconds	30s	780	Active scalping; Valtos recommended for Turns indicator	Good balance for scalpers. Enough ticks per bar for meaningful COT and imbalance detection. Valtos specifically mentions 30s for Turns.
45 seconds	45s	520	Scalping to short-term day trading	Uncommon. Most traders skip to 1min. Valid if 30s is too fast and 1min is too slow.

7.2.2 Minute-Based — Tier 2 (Most Common for Day Trading)

Bar Size	Config Value	Bars Per RTH Session	Typical Use Case	Order Flow Notes
1 minute	1m	390	Short-term day trading; delta divergence watch; Pulse signals on fast moves	Dale's preferred TF for watching CVD divergence. Enough volume for meaningful delta. Stacked imbalances become reliable.
2 minutes	2m	195	Day trading; bridge between 1min and 3min	Less common. Use when 1min is too choppy. Imbalances and COT very clear.

3 minutes	3m	130	Day trading; Pulse multi-TF confirmation (use as higher TF for 30s primary)	Valtos mentions 3-minute as the higher TF for Pulse multi-TF analysis when trading 30-second charts.
4 minutes	4m	97	Uncommon; transitional size	Rarely used in practice. Valid configuration.
5 minutes	5m	78	Primary intraday chart (most popular). Dale's preferred for VWAP analysis. Orderflows Pulse default.	Industry standard for order flow analysis. Dale explicitly states '5-minute is my personal preference for Daily VWAP trading.' Valtos uses 5min for many of his book examples. Best balance of signal quality vs noise.
6 minutes	6m	65	Uncommon; used by some prop traders	Valid but rare. Divides the RTH session into exact 65 bars.
10 minutes	10m	39	Medium-term day trading; swing trade entry timing	Fewer, cleaner bars. Stacked imbalance zones more meaningful. Good for higher-timeframe context.
12 minutes	12m	32	Uncommon	Valid. Used rarely.
15 minutes	15m	26	Stacked imbalance hunting (Valtos recommendation); swing trade context; Orderflows Inner Circle webinar examples	Valtos explicitly recommends 15-minute charts for finding stacked imbalances: 'I like to look for these trades on longer time frame charts like a 15 minute or 30 minute chart.' Strong institutional signals.
20 minutes	20m	19	Intermediate term; pre-market analysis	Clear, high-conviction bars. HVN and POC levels very significant at this TF.
30 minutes	30m	13	Swing trade context; Dale's preferred for Weekly VWAP; session structure analysis	Dale explicitly: 'for intraday trading with a 30-minute time frame... I find it more effective than the Daily VWAP.' Very clean Volume Profile and VWAP signals.
45 minutes	45m	8	Near-session-level analysis	Uncommon but valid. 8 bars per session gives macro view.
60 minutes / 1 hour	60m or 1h	6	Session structure; institutional bias; swing trade entry	6 bars per RTH session. Each bar = 1 hour of institutional activity. Very clear initiative vs responsive

				context.
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7.2.3 Hour and Session Based — Tier 3

Bar Size	Config Value	Bars Per Session / Day	Typical Use Case	Order Flow Notes
2 hours	2h	3 per RTH session	Macro intraday bias; open/midday/close structure	Each bar represents a major session segment. Delta and Volume Profile per segment shows institutional rotation.
3 hours	3h	2 per RTH session	Open vs afternoon analysis	2 bars: open drive + afternoon. Clear initiative vs responsive separation.
4 hours	4h	~1.6 per RTH session	Swing trading; daily bias chart	Standard for swing traders alongside daily chart.
6 hours	6h	1 per RTH session (approx)	Full-session single-bar; macro institutional view	Entire RTH session in one bar. Delta tells you if the day was net buying or selling.
Daily	1D	1 per calendar day	Swing trading; position trading; yearly VWAP analysis	One bar per day. Order flow shows daily institutional bias. Volume Profile becomes the session profile.
Weekly	1W	1 per 5 trading days	Position trading; long-term S/R identification	Weekly bars with order flow. Rare but powerful for identifying multi-week imbalance zones.
Monthly	1MO	~21 trading days	Long-term investment; yearly institutional mapping	Monthly footprints. Mostly for research and institutional-grade analysis.

7.3 RANGE-Based Bar Sizes

For RANGE bars, the user specifies the number of TICKS (minimum price increments) each bar spans. A new bar opens only when price moves outside the previous bar's range. Bars may take seconds or minutes to complete depending on volatility.

Instrument	Tick Size	Common Range Sizes	Notes
ES (E-mini S&P 500)	\$12.50 / 0.25pts	2, 4, 6, 8, 10, 12, 16, 20 ticks	Valtos uses 4-range as his default in many examples. 4 range = 1 point = \$50 per contract.
NQ (E-mini Nasdaq)	\$5.00 / 0.25pts	4, 6, 8, 10, 16, 20 ticks	NQ more volatile — use larger range than ES for equivalent signal quality.

CL (Crude Oil)	\$10.00 / 0.01	4, 6, 8, 10, 12 ticks	High volatility. Valtos teaches CL extensively in his courses.
GC (Gold)	\$10.00 / 0.10	4, 6, 8, 10 ticks	
ZB (30-yr T-Bond)	\$31.25 / 1/32	2, 4, 6, 8 ticks	Lower volatility instrument — smaller range sizes appropriate.
6E (Euro FX Futures)	\$12.50 / 0.0001	4, 6, 8, 10, 12 ticks	Dale's primary day-trading instrument. 6-range mentioned in his examples.
YM (Dow futures)	\$5.00 / 1pt	4, 6, 8, 10, 16 ticks	

***i** Range bars are Valtos's first choice because: (1) bars only form when price MOVES — eliminating time-wasting quiet periods, (2) each bar has a guaranteed price range making imbalance comparison consistent across bars, (3) order flow signals are cleaner with less noise from low-activity periods.*

7.4 VOLUME-Based Bar Sizes

For VOLUME bars, the user specifies the number of contracts that must trade before a new bar closes. Each bar represents approximately equal market participation regardless of time elapsed.

Volume Size	Config Value	When to Use	Notes
100 contracts	100	Thin markets; educational / development testing	Very small — many bars per minute in active markets. Not recommended for live trading on ES.
250 contracts	250	Thin futures markets (ZB, 6J)	
500 contracts	500	Moderately active instruments	
1,000 contracts	1000	ES, NQ standard scalping TF	1000-volume bars on ES = roughly 1 bar per 10–30 seconds during active trading.
2,500 contracts	2500	ES active day trading	Dale uses 2500-volume bars in some of his examples. Good balance.
5,000 contracts	5000	ES primary intraday — equivalent to roughly 5-minute time	More stable bar duration. Imbalances very reliable.
10,000 contracts	10000	ES — Valtos mentions 10000 volume chart in book	Valtos explicitly references the 10,000 volume E-mini SP chart. Each bar represents significant

			institutional participation.
25,000 contracts	25000	ES — session-level macro view	Very few bars per session. Each bar tells a meaningful institutional story.
Custom	any int	User-defined to match their instrument's typical session volume ÷ desired bar count	Formula: $\text{daily_avg_volume} \div \text{target_bars_per_session} = \text{volume_size}$

7.5 Multi-Chart Configuration (Recommended Workspace)

Valtos and Dale both recommend using MULTIPLE chart types simultaneously. The engine must support opening multiple bar streams per instrument. Dale's documented workspace:

Chart Position	Bar Type	Bar Size	Purpose in Workspace
Top Left — Context chart	TIME	30 min	Bigger picture. Bid+Ask total volume cells. Session structure, Volume Profile, Weekly VWAP context.
Top Right — Primary trading chart	RANGE or TIME	4-range or 5 min	Main execution chart. Full footprint with bid/ask split, imbalances, COT, delta.
Bottom Left — Confirmation chart	VOLUME or TIME	2500 vol or 1 min	CVD divergence watch. Delta trend. Short-term momentum confirmation.
Bottom Right — Tick / DOM chart	TIME	1 min or tick	Entry precision. Live delta building. Large print detection (Trades Filter).

***i** All four charts in the workspace are linked by instrument but NOT by bar type or size. Changing the instrument on one chart updates all four. The engine streams four independent bar type/size combinations simultaneously per instrument.*

7.6 Bar Configuration API Parameters

The `bar_type` and `bar_size` parameters apply across ALL API endpoints and WebSocket subscriptions:

Parameter	Type	Valid Values	Default	Notes
<code>bar_type</code>	enum string	TIME, RANGE, VOLUME	TIME	Case-insensitive. Also accepts T, R, V as

				shorthand.
bar_size	string or int	See §7.2–7.4 tables. Examples: 5s, 30s, 1m, 5m, 15m, 30m, 1h, 4h, 1D, 1W, 4 (range ticks), 5000 (volume contracts)	5m	For TIME: use duration string. For RANGE/VOLUME: use integer.
session_filter	enum string	ALL, RTH_ONLY, ETH_ONLY, GLOBEX	RTH_ONLY	RTH = Regular Trading Hours. ETH = Extended. GLOBEX = full 23h session for CME futures.
timezone	string	Any IANA timezone. e.g. America/New_York, UTC, Asia/Tokyo	UTC	Affects how session_open/close shorthands resolve. Does NOT change ts field format (always UTC ns).
include_incomplete_bar	bool	true / false	false	If true, the current open (building) bar is included in historical queries. Partial data — use with caution.
align_to_session	bool	true / false	true	If true, bar boundaries align to session open (9:30:00 ET for ES). If false, bars start from first tick of history.

7.7 Bar Size Validation Rules

The engine enforces the following validation rules on bar_size to prevent misconfiguration:

Rule	Constraint	Error if Violated
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TIME minimum	5 seconds (5s)	HTTP 400: bar_size_too_small. Sub-5s bars generate excessive data with minimal signal value.
TIME maximum	Monthly (1MO)	HTTP 400: bar_size_too_large for TIME type.
TIME — must be clean divisor of 1 hour	Valid: 5m, 6m, 10m, 12m, 15m, 20m, 30m, 60m. Invalid: 7m, 11m, 13m, 17m	HTTP 400: bar_size_not_clean_divisor. Irregular sizes create misaligned bar boundaries that confuse session analysis.
RANGE minimum	1 tick	HTTP 400: range_too_small
RANGE maximum	500 ticks	HTTP 400: range_too_large. Bars would rarely close.
VOLUME minimum	50 contracts	HTTP 400: volume_too_small
VOLUME maximum	1,000,000 contracts	HTTP 400: volume_too_large. Bars would rarely close.
RANGE on TIME chart	Not combinable	HTTP 400: bar_type_size_mismatch. Range size must use bar_type=RANGE.
Tick chart (non-standard)	Not supported in v1.0	HTTP 400: bar_type_not_supported. Tick charts (1 bar per trade) planned for v2.0.

7.8 Effect of Bar Size on Order Flow Signal Quality

Bar size has a direct impact on the quality and reliability of each order flow signal. This table guides developers in selecting appropriate bar sizes for each signal type:

Signal / Indicator	Too Small (Noisy)	Optimal Range	Too Large (Sparse)	Recommended Default
Footprint Imbalance	< 5s or < 2-range	30s–5m TIME, 4–8 RANGE, 1000–5000 VOL	> 30m TIME	5m TIME or 4-RANGE
Stacked Imbalance	< 1m TIME	5m, 15m, 30m TIME	> 1h TIME	15m TIME (Valtos recommendation)
Delta / CVD	< 5s	1m–5m TIME for intraday	> 30m TIME	1m for CVD watch, 5m for signal

CVD Divergence	< 1m TIME	1m–5m TIME	> 15m TIME	1m TIME
Orderflows Pulse	< 30s	30s–5m TIME, 4–8 RANGE	> 15m TIME	5m TIME (Dale default)
Orderflows Turns	< 15s	30s–3m TIME, 4-RANGE	> 10m TIME	30s TIME or 4-RANGE (Valtos recommendation)
Orderflows Ratio (Big Buyer/Seller)	< 30s	1m–15m TIME	> 1h TIME	5m TIME
Single Prints (Last Buyer/Seller)	< 30s	1m–30m TIME	> 2h TIME	5m–15m TIME
Volume Profile POC/VAH/VAL	N/A — VP is always accurate	5m–30m TIME for clear profile	N/A	5m or 30m TIME
Volume Profile Shape (D/P/b/Thin)	< 5m TIME (profile not mature)	30m–1h TIME (profile mature mid-session)	N/A — shape only meaningful for full session	Check shape after 2 hours of RTH trading
VWAP Reaction Signal	< 30s (too many false touches)	1m–5m TIME	> 30m TIME	5m TIME (Dale explicit)
COT (Commitment of Traders)	< 1m TIME	5m–15m TIME	> 1h TIME	5m TIME
Absorption Detection	< 1m TIME (not enough volume)	5m–30m TIME	> 2h TIME	5m–15m TIME
Market Sweep Detector	< 15s	30s–5m TIME, 4–6 RANGE	> 30m TIME	1m TIME or 4-RANGE
Trapped Buyers/Sellers	< 1m TIME	5m–30m TIME	> 1h TIME	5m–15m TIME

 Signals generated on bars smaller than their 'Too Small' threshold will be flagged in the API response with `signal_quality=LOW`. Developers should filter these out or treat them as advisory only.

7.9 Bar Size in WebSocket Subscriptions

When subscribing to live streams, `bar_type` and `bar_size` are encoded in the topic string:

```
SUBSCRIBE footprint:ES:TIME:5m // 5-minute TIME bars for ES
```



```

SUBSCRIBE footprint:ES:RANGE:4           // 4-tick RANGE bars for ES
SUBSCRIBE footprint:ES:VOLUME:5000       // 5000-contract VOLUME bars for ES
SUBSCRIBE delta:live:ES:TIME:1m          // live delta on 1-minute bars
SUBSCRIBE pulse:ES:TIME:5m               // Pulse signals on 5-minute bars
SUBSCRIBE turns:ES:RANGE:4               // Turns signals on 4-range bars
SUBSCRIBE imbalances:ES:TIME:15m         // Imbalances on 15-minute bars
SUBSCRIBE profile:live:ES:TIME:5m        // Live session Volume Profile, 5-min
resolution

```

i A developer can subscribe to **MULTIPLE** bar sizes simultaneously for the same instrument. Example: subscribe to `pulse:ES:TIME:5m` AND `pulse:ES:TIME:30m` to implement multi-timeframe confluence detection in their own code.

7.10 Bar Size in REST Historical Queries

All historical endpoints accept `bar_type` and `bar_size` as query parameters:

```

GET /v1/footprint/ES?bar_type=TIME&bar_size=5m&from_ts=session_open&to_ts=now
GET /v1/footprint/ES?bar_type=RANGE&bar_size=4&from_ts=2026-05-30T09:30:00Z
GET /v1/signals/pulse/ES?bar_type=TIME&bar_size=5m&from_ts=now-5d&to_ts=now
GET /v1/signals/stacked_imbalance/ES?bar_type=TIME&bar_size=15m&from_ts=now-30d
GET /v1/profile/ES/session?bar_type=TIME&bar_size=5m&session_date=2026-05-30

```

i Historical data is stored at the **TICK** level and re-aggregated into any requested `bar_type`/`bar_size` on the fly. This means a developer can query any bar size from history without that exact size having been pre-computed. Re-aggregation adds ~50–200ms to query time for large date ranges.

8. Timestamp Specification — Signal vs Snapshot

8.1 The Fundamental Difference

Timestamp Type	One-Line Definition	Triggered By	Trading Significance
Signal Timestamp	The exact nanosecond a specific market condition BECAME TRUE	A market event — a condition crossing a threshold for the first time	HIGH — marks the birth of an actionable setup
Snapshot Timestamp	The exact nanosecond a periodic reading of an indicator was RECORDED	A timer or bar close — regardless of whether the value changed	LOW — reflects current state, not a new event

8.2 Side-by-Side Architecture

Property	Signal Timestamp	Snapshot Timestamp
Subsystem	Signal Detection Engine — event-driven	Metric Streaming Engine — timer-driven
Trigger	Condition threshold crossed for first time	Timer fires (100ms, 5s) OR bar closes
Frequency	Sparse — minutes or hours between signals on same instrument	Dense — tens to hundreds per minute
Deduplication	YES — same signal not re-emitted until condition resets	NO — snapshots repeat even if value unchanged
Value change required?	YES — fires only when something NEW happens	NO — emits even if value is identical to last snapshot
Use in API query	Filter: 'show me signals BETWEEN these two times'	Filter: 'show me the indicator VALUE AT this time'
Persisted in DB?	YES — every signal permanently stored with full context	NO — latest in Redis cache; history via bar OHLCV
WebSocket behavior	Fires once per event. Quiet between events.	Fires on every timer tick. Continuous stream.

8.3 The Four Signal Timestamp Fields

Every signal record contains four timestamp fields — each with a distinct meaning:

Field	What It Records	When to Use
-------	-----------------	-------------

detection_ts	Nanosecond the engine DETERMINED the condition was met. Always after bar close for bar-close signals. Primary signal timestamp.	Strategy logic. 'When did I know about this signal?' Use in backtesting as the action timestamp.
bar_open_ts	Nanosecond the BAR that triggered the signal OPENED. Always earlier than detection_ts.	Bar attribution. 'Which bar caused this?' Use to align signal with chart data.
bar_close_ts	Nanosecond the triggering bar CLOSED. For bar-close signals ≈ detection_ts.	Execution timing. 'Enter at open of the bar AFTER this timestamp.'
exchange_ts	Exchange-assigned timestamp of the LAST TICK that completed the condition. Most precise market-event time.	Regulatory / audit use. Matches exchange records exactly.

These four always follow this strict ordering:

```
exchange_ts <= bar_open_ts < bar_close_ts <= detection_ts
```

8.4 All Valid Timestamp Formats

Format	Example	Notes
Unix nanoseconds (int64)	1748599800000000000	Preferred. 19 digits. Unambiguous. Fastest to parse. Use in all production code.
Unix milliseconds (int64)	1748599800000	13 digits. Auto-detected by digit count. Converted internally to ns (× 1,000,000).
Unix seconds (int64 or float64)	1748599800 or 1748599800.291	10 digits integer or float with fractional seconds.
ISO 8601 UTC	2026-05-30T09:35:00Z	Z suffix REQUIRED. Without it: HTTP 400 missing_timezone.
ISO 8601 UTC with nanoseconds	2026-05-30T09:35:00.000003847Z	Up to 9 decimal places accepted.
ISO 8601 with timezone offset	2026-05-30T04:35:00-05:00	Offset accepted. Internally converted to UTC.
Date-only string	2026-05-30	Expands to 2026-05-30T00:00:00.000000000Z (midnight UTC).
Relative: now	now	Current server UTC time at query execution.
Relative: now-Xs	now-30s, now-45s	X seconds ago. Min 1s.
Relative: now-Xm	now-15m, now-30m	X minutes ago.

Relative: now-Xh	now-4h, now-6h	X hours ago.
Relative: now-Xd	now-1d, now-5d	X calendar days ago.
Relative: now-Xw	now-2w, now-4w	X weeks ago.
Relative: now-XM	now-3M, now-6M	X months ago (capital M = months).
Relative: now-Xy	now-1y, now-2y	X years ago.
session_open	session_open	Today's RTH session open for the queried instrument (e.g. 09:30:00 ET for ES).
session_close	session_close	Today's RTH session close (e.g. 16:00:00 ET for ES).
prev_session_open	prev_session_open	Previous trading day's session open.
prev_session_close	prev_session_close	Previous trading day's session close.
week_open	week_open	Monday's session open for the current trading week.
month_open	month_open	First trading day of the current calendar month, session open.
year_open	year_open	First trading day of current calendar year, session open.
epoch	epoch	2010-01-01T00:00:00Z — fetch the maximum available history.

8.5 Valid vs Invalid — Quick Examples

✓ 1748599800000000000 — Unix nanoseconds, 19 digits, within range
 ✓ 2026-05-30T09:35:00Z — ISO 8601 UTC with Z suffix
 ✓ 2026-05-30T04:35:00-05:00 — ISO with offset (same UTC moment)
 ✓ now-4h — valid relative shorthand
 ✓ session_open — session-relative shorthand
 ✓ 1748599800.291 — Unix seconds with fractional part
 ✓ epoch — fetch all history

✗ 2026-05-30T09:35:00 — INVALID: missing timezone suffix → HTTP 400 missing_timezone
 ✗ 05/30/2026 09:35:00 — INVALID: US date format not supported → HTTP 400 invalid_timestamp_format
 ✗ now+1d — INVALID: future timestamps not allowed → HTTP 400 timestamp_in_future
 ✗ 0 — INVALID: Unix epoch zero predates all data → HTTP 400 timestamp_too_early
 ✗ 1748599800000 — AMBIGUOUS: 13 digits treated as milliseconds, not nanoseconds — be explicit

8.6 Signal Timestamp Nuances by Signal Type

Signal Type	detection_ts Behavior	Important Developer Note
Stacked Imbalance	Set when Nth consecutive imbalance is detected at bar close	zone_id persists. Extensions don't generate new detection_ts. Use zone_created_ts vs zone_last_updated_ts.
Orderflows Pulse	Set at bar close when composite score first crosses threshold	Can re-fire: if score drops then rises again, new detection_ts on new bar.
Orderflows Turns	Set at TICK level when turning pattern completes — not bar close	exchange_ts ≈ detection_ts within microseconds. Most precise signal.
Orderflows Ratio	Set at bar close when bid/ask ratio at extreme crosses threshold	One signal per bar maximum. Consecutive qualifying bars each get own detection_ts.
Single Print	Set at bar close when zero/near-zero volume at extreme detected	magnet_active=true persists until resolved. magnet_deactivation_ts records resolution.
Delta Surge	Set when surge_magnitude first crosses threshold (every 100ms)	detection_ts = first crossing. surge_peak_ts = when surge peaked.
CVD Divergence	Set when divergence confirmed for N consecutive bars	detection_ts = Nth bar confirmation. divergence_start_ts = when pattern began.
VWAP Reaction	Set at TICK level when price touches VWAP and order flow confirms	Mid-bar signal — entry may be possible on same bar. Use exchange_ts for precise entry timing.
Volume Profile Shape	Set at each bar close as shape evolves during session build	Check shape_confidence field. Low confidence early in session — shape may change.

8.7 Timestamp Query Quick Reference

I want to...	Parameter	Example
Get signals today	from_ts=session_open	GET /v1/signals/pulse/ES? from_ts=session_open
Get signals in specific hour	from_ts + to_ts	?from_ts=2026-05-30T14:00:00Z&to_ts=2026-05-30T15:00:00Z
Identify which bar caused signal	bar_close_ts field	Read bar_close_ts from signal response
Find entry timing after signal	bar_close_ts + 1 bar	Enter at open of bar AFTER bar_close_ts

Get current VWAP	latest=true	GET /v1/vwap/ES?latest=true
Get VWAP at specific moment	at_ts	GET /v1/vwap/ES?at_ts=2026-05-30T14:00:00Z
Run backtest last 6 months	from_ts=now-6M	POST /v1/backtest { from_date: 'now-6M', to_date: 'now' }
Get only RTH session signals	from_ts=session_open&to_ts=session_close	Excludes pre/post market
Get unresolved Single Print magnets	resolved=false	GET /v1/signals/single_prints/ES?resolved=false
Get last 5 Pulse signals	order+limit	? order_by=detection_ts&direction=desc&limit=5
Replay missed signals on WS connect	since=now-1h	SUBSCRIBE pulse:ES?since=now-1h

9. Tick Classification — Lee-Ready Algorithm

Every order flow metric begins with tick classification: determining whether each trade was initiated by an aggressive buyer (ask-side) or aggressive seller (bid-side). The Lee-Ready Algorithm (Lee & Ready, Journal of Finance, 1991) is the industry standard.

1.1 Lee-Ready Algorithm — Step-by-Step

Input Variables

P_t	= Price of the current trade
M_t Ask _t / 2	= Midpoint of prevailing bid-ask spread = (Bid _t +
Bid_t	= Best bid price at time of trade
Ask_t	= Best ask price at time of trade
P_{t-1}	= Price of the immediately preceding trade

Rule 1 — Quote Rule (applied first)

```
IF Pt > Mt → CLASSIFY as ASK (aggressive buyer)
IF Pt < Mt → CLASSIFY as BID (aggressive seller)
IF Pt = Mt → INDETERMINATE → apply Rule 2
```

Rule 2 — Tick Rule (applied when Quote Rule is indeterminate)

```
IF Pt > P{t-1} → CLASSIFY as ASK (uptick = buyer initiated)
IF Pt < P{t-1} → CLASSIFY as BID (downtick = seller initiated)
IF Pt = P{t-1} → CLASSIFY as LAST (zero-tick = carry forward
previous classification)
```

IQFeed Direct Aggressor Override (CME & ICE only)

```
IF TradeAggressor_field = 1 → CLASSIFY as ASK (exchange-
confirmed buyer)
IF TradeAggressor_field = 2 → CLASSIFY as BID (exchange-
confirmed seller)
IF TradeAggressor_field = 0 → Apply Lee-Ready above
```

★ Exchange-direct aggressor classification (IQFeed, CME/ICE) achieves 100% accuracy. Lee-Ready achieves ~85-99% accuracy depending on market conditions. Always prefer exchange-direct when available.

Result Mapping to Volume Buckets

```
Classification = ASK → ask_vol[price] += trade_volume
Classification = BID → bid_vol[price] += trade_volume
total_vol[price]    = ask_vol[price] + bid_vol[price]
```


10. Delta Calculations

Delta is the net difference between aggressive buying volume (ask-side) and aggressive selling volume (bid-side). It is the most fundamental order flow metric.

2.1 Bar Delta

Formula

```
BarDelta = SUM(ask_vol[i]) - SUM(bid_vol[i])
          = TotalAskVol_bar - TotalBidVol_bar
```

Where the sum is over all price levels *i* within the bar.

BarDelta
net buying, Negative = net selling.

= Net aggressive buying pressure for one bar. Positive =

ask_vol[i]

= Ask-side (aggressive buyer) volume at price level *i*

bid_vol[i]

= Bid-side (aggressive seller) volume at price level *i*

Worked Example (5-minute ES bar)

Price Level	bid_vol	ask_vol	Level Delta
4502.25	120	180	+60
4502.50	200	150	-50
4502.75	90	310	+220
4503.00	300	100	-200
TOTAL	710	740	BarDelta = 740 - 710 = +30

2.2 Max Delta & Min Delta

Formula

```
MaxDelta = max(RunningDelta_t) for all ticks t within the bar
```

```
MinDelta = min(RunningDelta_t) for all ticks t within the bar
```

```
RunningDelta_t = SUM of per-tick delta from bar_open to tick t
                = RunningDelta_{t-1} + delta(tick_t)
```

```
delta(tick_t) = +volume if tick classified as ASK
```

```
                = -volume if tick classified as BID
```

★ *MaxDelta and MinDelta track the peak and trough of the running delta within the bar — measuring how far the delta swung in each direction before the bar closed. A bar with MaxDelta=+500 and BarDelta=+50 tells a very different story than a bar with MaxDelta=+500 and BarDelta=+490.*

2.3 Delta Percentage (Delta/Volume %)

Formula

$$\text{DeltaPct} = (\text{BarDelta} / \text{TotalBarVolume}) \times 100$$

$$\begin{aligned} \text{TotalBarVolume} &= \text{SUM}(\text{ask_vol}[i]) + \text{SUM}(\text{bid_vol}[i]) \quad \text{for all } i \\ &= \text{TotalAskVol_bar} + \text{TotalBidVol_bar} \end{aligned}$$

Example: BarDelta = +30, TotalBarVolume = 1450 → DeltaPct = (30/1450) × 100 = +2.07%

★ *DeltaPct normalises delta for comparison across bars of different sizes. A delta of +30 on a 100-contract bar is more significant than +30 on a 10,000-contract bar. DeltaPct makes these comparable.*

2.4 Cumulative Delta (CVD)

Formula

$$\begin{aligned} \text{CVD}_n &= \text{SUM}(\text{BarDelta}_i) \quad \text{for } i = 1 \text{ to } n \\ &= \text{CVD}_{\{n-1\}} + \text{BarDelta}_n \end{aligned}$$

CVD resets to 0 at each session open (session_open_ts).

$$\text{CVD}_1 = \text{BarDelta}_1 \quad (\text{first bar of session})$$

CVD_n = Cumulative Delta after bar n. Running session total.

BarDelta_n = Delta of bar n

Example session: Bar1 delta=+150, Bar2 delta=-80, Bar3 delta=+220 → CVD after Bar3 = 150 + (-80) + 220 = +290

★ *CVD divergence: if Price makes a new high while CVD makes a lower high, buying pressure is weakening — potential reversal signal. If Price makes a new low while CVD makes a higher low, selling pressure is weakening — potential bounce signal.*

2.5 Delta Divergence Detection

Standard Divergence Formula

BullishDivergence = $\text{Price}_n < \text{Price}_{\{n-k\}}$ AND $\text{CVD}_n > \text{CVD}_{\{n-k\}}$
BearishDivergence = $\text{Price}_n > \text{Price}_{\{n-k\}}$ AND $\text{CVD}_n < \text{CVD}_{\{n-k\}}$

Where k = lookback period (default $k = 1$ to 5 bars)

Price_n = close price of bar n

CVD_n = cumulative delta at bar n close

Bar-Level Delta Divergence (Valtos Webinar 18)

BarDeltaDivergence_Bullish = $\text{BarClose} < \text{BarOpen}$ AND $\text{BarDelta} > 0$
 (Bearish candle + positive delta → sell pressure failed to close price lower)

BarDeltaDivergence_Bearish = $\text{BarClose} > \text{BarOpen}$ AND $\text{BarDelta} < 0$
 (Bullish candle + negative delta → buy pressure failed to close price higher)

★ *Bar-level delta divergence is a micro-divergence within a single bar. Trade in the direction of the candle color — the candle won despite adverse delta, showing hidden strength.*

11. Imbalance Detection

Imbalances occur when aggressive volume on one side of the market significantly exceeds the opposing side at a price level. The diagonal comparison is the defining characteristic of footprint chart imbalance detection.

3.1 Single Buying Imbalance

Formula — Diagonal Comparison

BuyingImbalance at price P = TRUE

IF ask_vol[P] >= ImbalanceRatio x bid_vol[P - 1_tick]
AND bid_vol[P - 1_tick] > 0 (avoid division by zero)

Where:

ask_vol[P] = ask-side volume at current price level P
 bid_vol[P-1tick] = bid-side volume at ONE TICK BELOW P
 ImbalanceRatio = user-configurable threshold (default 3.0 = 300%)

Formula — Single Selling Imbalance

SellingImbalance at price P = TRUE

IF bid_vol[P] >= ImbalanceRatio x ask_vol[P + 1_tick]
AND ask_vol[P + 1_tick] > 0 (avoid division by zero)

Where:

bid_vol[P] = bid-side volume at current price P
 ask_vol[P+1tick] = ask-side volume at ONE TICK ABOVE P

 The comparison is **DIAGONAL** — not horizontal. ask_vol at price P is compared against bid_vol at price P-1_tick, NOT against bid_vol at price P. This diagonal comparison is what distinguishes footprint imbalance from simple volume dominance.

Common Ratio Thresholds

Ratio Setting	Sensitivity	Typical Use
2.5x (250%)	High — detects more imbalances	Scalping on 30s or 1min bars; active markets
3.0x (300%)	Standard — industry default	5-minute bars; balanced signal quality
4.0x (400%)	Low — only strong imbalances	15-minute or 30-minute bars; high-conviction only

5.0x (500%)	Very selective	Macro-level analysis; daily/weekly context only
-------------	----------------	---

3.2 Stacked Imbalance

Formula

StackedBuyImbalance = TRUE

IF consecutive count of **BuyingImbalance = TRUE**
at vertically adjacent price levels **>= StackedMinCount**

i.e. **BuyingImbalance[P] = TRUE**
BuyingImbalance[P+1] = TRUE
BuyingImbalance[P+2] = TRUE (minimum 3, default)
 ...

ZoneHigh = highest price level in the consecutive imbalance chain

ZoneLow = lowest price level in the consecutive imbalance chain

ZoneStrength = count of consecutive imbalances x
avg(**ImbalanceRatio**)

ImbalanceRatio_avg = MEAN of actual ratios at each level in the stack

★ *Stacked imbalances persist across bars as support/resistance zones. Zone_ID is assigned on first detection. The zone expands if subsequent bars add more imbalances at adjacent levels. The zone is resolved (deactivated) when price trades through it.*

3.3 Imbalance Ratio (Actual)

Formula

ActualRatio_Buy[P] = ask_vol[P] / bid_vol[P - 1_tick]

ActualRatio_Sell[P] = bid_vol[P] / ask_vol[P + 1_tick]

ImbalanceStrengthScore = CLAMP((ActualRatio / ImbalanceRatio) x 5, 1, 10)

Where **CLAMP(x, min, max)** limits the value to [min, max] range.

Example: ImbalanceRatio=3.0, ActualRatio=7.2 → StrengthScore = CLAMP((7.2/3.0)×5, 1, 10) = CLAMP(12.0, 1, 10) = 10 (maximum strength)

3.4 Absorption Detection

Formula

Absorption at price level P = TRUE

IF total_vol[P] >= AbsorptionVolThreshold
AND |delta[P]| <= AbsorptionDeltaRatio x total_vol[P]

i.e. total_vol[P] >= 500 (default threshold: 500 contracts)
 |ask_vol[P] - bid_vol[P]| <= 0.10 x total_vol[P]
 (net delta is less than 10% of total volume – near-zero net direction)

AbsorptionDeltaRatio = user-configurable (default 0.10 = 10%)

total_vol[P]	= ask_vol[P] + bid_vol[P] — total volume at the level
 delta[P] 	= absolute value of ask_vol[P] - bid_vol[P]

★ Absorption requires HIGH volume AND near-zero net delta. This distinguishes it from consolidation (low volume AND near-zero delta). The high volume proves institutional inventory transfer is occurring.

12. Volume Profile Calculations

4.1 Profile Construction

Algorithm

1. Collect all ticks in the profile period [T_start, T_end]
2. For each tick:
 - profile[price_bucket] += tick_volume
 - where price_bucket = ROUND(tick_price / tick_size) x tick_size
3. result: profile[] = histogram of volume at each price level

Price Bucket Width

```

tick_size = minimum price increment of the instrument
           = 0.25 pts for ES (= $12.50/contract)
           = 0.01 for NYSE stocks
           = 0.0001 for FX pairs

```

profile[bucket] accumulates ALL volume traded within that tick_size increment.

4.2 Point of Control (POC)

Formula

```

POC = argmax(profile[price])
     = price level with the HIGHEST total_vol in the profile period

```

```

POC_volume = max(profile[price]) for all prices in period

```

If two price levels have equal volume:

```

POC = lower of the two prices (conservative, closer to value)

```

4.3 Value Area Calculation

Algorithm (Industry Standard — Two-Price-Level Expansion)

Step 1: `target_vol = TotalProfileVolume x ValueAreaPct (default 0.70)`

Step 2: `accumulated_vol = profile[POC]`
`VA_high = POC`
`VA_low = POC`

Step 3: `WHILE accumulated_vol < target_vol:`

`upper_2 = profile[VA_high + 1_tick] + profile[VA_high + 2_tick]`
`lower_2 = profile[VA_low - 1_tick] + profile[VA_low - 2_tick]`

`IF upper_2 >= lower_2:`
`VA_high += 2_ticks`
`accumulated_vol += upper_2`
`ELSE:`
`VA_low -= 2_ticks`
`accumulated_vol += lower_2`

Step 4: `VAH = VA_high`
`VAL = VA_low`
`VA%_actual = accumulated_vol / TotalProfileVolume x 100`

★ *The two-price-level expansion (comparing 2 levels above vs 2 levels below) is the industry standard used by Trading Technologies X_STUDY, Sierra Chart, and NinjaTrader. Some platforms expand one level at a time — results will differ slightly.*

Worked Example

Profile total volume = 10,000 contracts. Target = 7,000 (70%). POC volume = 2,100.

Iteration 1: `upper_2=800+600=1400`, `lower_2=400+500=900` → expand up. VA: [POC, POC+2]. Accumulated=3500.

Iteration 2: `upper_2=500+300=800`, `lower_2=900+700=1600` → expand down. VA: [POC-2, POC+2]. Accumulated=5100.

Continue until accumulated ≥ 7000. Final VAH and VAL define the value area.

4.4 High Volume Node (HVN) Detection

Formula

`mean_vol = TotalProfileVolume / NumberOfPriceLevels`


```

HVN[price] = TRUE IF profile[price] >= HVNMultiplier x
mean_vol

HVNMultiplier = user-configurable (default 1.5 = 150% of mean)

HVN Strength Score = profile[price] / mean_vol

```

4.5 Low Volume Node (LVN) Detection

Formula

```

LVN[price] = TRUE IF profile[price] <= LVNMultiplier x
mean_vol

LVNMultiplier = user-configurable (default 0.5 = 50% of mean)

LVN Gap = consecutive price levels where LVN[price] = TRUE
LVN Width = number of consecutive LVN levels

```

4.6 Volume Profile Shape Classification

Formula

```

Define:
    profile_mean    = SUM(price[i] x profile[i]) / TotalProfileVolume
    profile_stdev   = SQRT( SUM((price[i] - mean)^2 x profile[i]) /
TotalVol )
    poc_percentile  = (POC - ProfileLow) / (ProfileHigh - ProfileLow)
    tail_pct_lower  = volume below (POC - 1.5 x profile_stdev) /
TotalVol
    tail_pct_upper  = volume above (POC + 1.5 x profile_stdev) /
TotalVol

D-Profile    = poc_percentile in [0.35, 0.65] AND profile_stdev <
threshold
P-Profile    = poc_percentile > 0.60 AND tail_pct_lower > 0.15
b-Profile    = poc_percentile < 0.40 AND tail_pct_upper > 0.15
Thin-Profile  = profile_stdev > 2.0 x mean_stdev_for_this_instrument

```

13. VWAP — Volume Weighted Average Price

5.1 Core VWAP Formula

Formula

$$\begin{aligned}\text{VWAP}_n &= \text{SUM}(\text{Price}_i \times \text{Volume}_i) / \text{SUM}(\text{Volume}_i) \\ &= \text{CumulativePriceVolume}_n / \text{CumulativeVolume}_n\end{aligned}$$

Where $i = 1$ to n (all trades from anchor point to current tick n)

Incremental update (efficient – no full recalculation each tick):

$$\begin{aligned}\text{CumPV}_n &= \text{CumPV}_{\{n-1\}} + (\text{Price}_n \times \text{Volume}_n) \\ \text{CumVol}_n &= \text{CumVol}_{\{n-1\}} + \text{Volume}_n \\ \text{VWAP}_n &= \text{CumPV}_n / \text{CumVol}_n\end{aligned}$$

Price_i = Typical Price of trade i = Last traded price (tick-level VWAP)

Volume_i = Number of contracts/shares at trade i

★ For bar-level VWAP (when using bar data rather than tick data): Typical Price = (High + Low + Close) / 3. The OFE engine uses tick-level prices for maximum precision since tick data is available.

5.2 VWAP Anchor Types

Anchor Type	Anchor Point (T_start)	Reset Behavior
Daily VWAP	Session open time (e.g. 09:30:00 ET for ES)	Resets each session open
Weekly VWAP	Monday session open time	Resets each Monday
Monthly VWAP	First trading day of month, session open	Resets each month
Yearly VWAP	First trading day of year, session open	Resets each January
Anchored VWAP	User-selected timestamp or price event	Does not auto-reset — persists until removed
Session VWAP	Same as Daily VWAP for equities; Globex open for futures	Configurable per instrument

Anchored VWAP formula is identical to core VWAP:

$$\text{VWAP}_{\text{anchored}}(\text{T_anchor}, t) = \text{SUM}(\text{P}_i \times \text{V}_i) / \text{SUM}(\text{V}_i)$$

for all i where $\text{T_anchor} \leq \text{timestamp}_i \leq t$

5.3 VWAP Standard Deviation Bands

Formula

Step 1: Calculate variance at each tick n:

$$\text{Variance}_n = \text{CumPV2}_n / \text{CumVol}_n - \text{VWAP}_n^2$$

$$\text{CumPV2}_n = \text{CumPV2}_{\{n-1\}} + (\text{Price}_n^2 \times \text{Volume}_n)$$

(tracks cumulative sum of price-squared weighted by volume)

Step 2: Standard Deviation:

$$\text{StdDev}_n = \text{SQRT}(\text{Variance}_n)$$

Step 3: Bands:

$$\text{Band}_{+k} = \text{VWAP}_n + (k \times \text{StdDev}_n) \quad \text{upper band, k-th deviation}$$

$$\text{Band}_{-k} = \text{VWAP}_n - (k \times \text{StdDev}_n) \quad \text{lower band, k-th deviation}$$

Standard values: k = 1, 2, 3

★ *This is a VOLUME-WEIGHTED standard deviation — not the same as simple standard deviation. It weights the variance by the volume traded at each price, making it more sensitive to high-volume price levels. This is what Dale's book refers to as the 'VWAP deviation band'.*

Example: VWAP=4502.50, StdDev=8.25 → Band+1=4510.75, Band-1=4494.25,
Band+2=4519.00, Band-2=4486.00

5.4 VWAP Signal Conditions

VWAP Reaction Signal

$$\text{VWAPTouch} = \text{ABS}(\text{CurrentPrice} - \text{VWAP}_n) \leq \text{VWAPTouchThreshold_ticks} \times \text{tick_size}$$

$$\begin{aligned} \text{VWAP_Reaction_Long} = & \text{VWAPTouch} = \text{TRUE} \\ & \text{AND CurrentPrice_prev} > \text{VWAP}_n \quad (\text{approaching} \\ & \text{from above}) \\ & \text{AND BarDelta} > 0 \quad (\text{positive delta} \\ & \text{confirmation}) \end{aligned}$$

```
VWAP_Reaction_Short = VWAPTouch = TRUE  
AND CurrentPrice_prev < VWAP_n (approaching  
from below)  
AND BarDelta < 0 (negative delta  
confirmation)
```

VWAP Rotation Signal

```
VWAP_Rotation_Long = CurrentPrice <= Band_-1  
AND BarDelta > 0 (buying at  
lower band)  
AND CVD divergence confirmed (optional)  
  
VWAP_Rotation_Short = CurrentPrice >= Band_+1  
AND BarDelta < 0 (selling at  
upper band)
```

14. Commitment of Traders (COT) — Intrabar

Note: This is Valto's proprietary intrabar COT concept — NOT the CFTC weekly COT report. It identifies the price level within a bar where institutional commitment was highest.

6.1 COT Price Formula

Formula

```

COT_price = argmax(total_vol[price]) within the current bar
           = price level with maximum ask_vol[P] + bid_vol[P]

COT_volume = max(total_vol[price]) within the current bar
            = max(ask_vol[P] + bid_vol[P]) for all P in bar

COT_delta  = ask_vol[COT_price] - bid_vol[COT_price]

```

COT Position Interpretation

```

COT_position_ratio = (COT_price - BarLow) / (BarHigh - BarLow)

IF COT_position_ratio < 0.33 (COT near bar low):
  AND BarClose > BarOpen (bullish bar):
    → Strong passive buyers at lows – bullish signal

IF COT_position_ratio > 0.67 (COT near bar high):
  AND BarClose < BarOpen (bearish bar):
    → Strong passive sellers at highs – bearish signal

IF COT_position_ratio in [0.33, 0.67] (COT in middle):
  → Neutral – two-sided market facilitation

```

15. Orderflows Pulse — Composite Score Formula

The Pulse is a 7-variable composite indicator. Each variable produces a sub-score from 0-100. The composite score is a weighted sum. A score ≥ 70 is considered high-conviction.

7.1 Variable Sub-Scores

Variable 1: Order Flow Score

```
OF_score = CLAMP(ABS(BarDelta) / BarVolume x 200, 0, 100)
```

Interpretation: measures how aggressively one side is dominating.
 0 = perfectly balanced. 100 = 50%+ delta/volume ratio (very aggressive).

Variable 2: Delta Score

```
Delta_score = CLAMP(ABS(BarDelta) / MaxDelta_session x 100, 0, 100)
```

Interpretation: how large is this bar's delta relative to today's maximum?
 100 = this is the most aggressive delta bar of the session.

Variable 3: POC Score

```
POC_position = ABS(BarClose - COT_price) / (BarHigh - BarLow)
POC_score = CLAMP((1 - POC_position) x 100, 0, 100)
```

Interpretation: how close is bar close to the bar's COT price?
 100 = bar closed exactly at the COT level (maximum institutional alignment).
 0 = bar closed far from the COT level.

Variable 4: Imbalance Score

```
ImbalanceCount = number of imbalance cells in this bar
Imbalance_score = CLAMP(ImbalanceCount / 5 x 100, 0, 100)
```

Interpretation: more imbalances = stronger imbalance score.
 Stacked imbalances get bonus: score x 1.5 if StackedCount $\geq$ StackedMinCount.

Variable 5: Volume Score

```

AvgBarVolume_session = CumVolume_session / BarCount_session
Volume_score = CLAMP(BarVolume / AvgBarVolume_session x 100, 0, 100)

```

Interpretation: is this bar's volume above average?

100 = bar volume is $\geq$ average (sufficient participation for valid signal).

Below average volume $\rightarrow$ score $< 100 \rightarrow$ reduces composite score.

Variable 6: Price Action Score

```

BarRange      = BarHigh - BarLow
AvgRange      = EMA(BarRange, 14)    (14-bar EMA of bar range)
CandleBody    = ABS(BarClose - BarOpen)
BodyRatio     = CandleBody / BarRange

```

```

PA_score = CLAMP(BodyRatio x (BarVolume / AvgBarVolume) x 100, 0, 100)

```

Interpretation: strong close relative to range + above-average volume = high PA score.

Variable 7: Swing Score

```

SwingHigh_N = highest bar high in last N bars (default N = 10)
SwingLow_N  = lowest bar low in last N bars

```

```

SwingPct = (BarClose - SwingLow_N) / (SwingHigh_N - SwingLow_N)

```

IF PulseDirection = LONG:

```

Swing_score = CLAMP(SwingPct x 100, 0, 100)
(100 = bar closed at top of recent swing range – momentum up)

```

IF PulseDirection = SHORT:

```

Swing_score = CLAMP((1 - SwingPct) x 100, 0, 100)
(100 = bar closed at bottom of recent swing range – momentum down)

```

7.2 Composite Pulse Score

Formula

Weights (default, sum to 1.0):

```

w_OF   = 0.20   (order flow dominance)
w_D    = 0.20   (delta strength)
w_POC  = 0.15   (POC alignment)
w_IMB  = 0.20   (imbalance count)
w_VOL  = 0.10   (volume filter)
w_PA   = 0.10   (price action)
w_SW   = 0.05   (swing context)

```

```

PulseScore = w_OF x OF_score
            + w_D  x Delta_score
            + w_POC x POC_score
            + w_IMB x Imbalance_score
            + w_VOL x Volume_score
            + w_PA  x PA_score
            + w_SW  x Swing_score

```

PulseScore range: 0 - 100

Signal threshold: PulseScore >= 70 → emit Pulse signal

Direction Determination

```

IF BarDelta > 0 AND BarClose > BarOpen:
    PulseDirection = LONG

```

```

IF BarDelta < 0 AND BarClose < BarOpen:
    PulseDirection = SHORT

```

```

IF signs conflict (BarDelta > 0 but BarClose < BarOpen):
    PulseDirection = NEUTRAL (Bar Delta Divergence – no Pulse
    signal)

```

Pulse signal only fires if PulseScore >= threshold AND Direction != NEUTRAL.

16. Orderflows Turns — Turning Point Score

The Turns indicator uses 6 variables to detect market turning points. Scores from 0-6 (one point per variable alignment). A score ≥ 3 is a valid Turn signal; ≥ 5 is high-conviction.

8.1 Six-Variable Turning Point Score

```
TurnScore = V1_poc + V2_bid_ask + V3_volume + V4_price_action +
V5_delta + V6_swing
```

Each variable contributes 1 point if its condition is met:

V1_poc (POC):

```
+1 IF ABS(BarHigh - COT_price) <= 2_ticks (turn at high,
bearish)
    OR ABS(BarLow - COT_price) <= 2_ticks (turn at low,
bullish)
```

V2_bid_ask (Volume at extremes):

```
+1 IF ask_vol[BarHigh_price] < VolumeExhaustThreshold
    OR bid_vol[BarLow_price] < VolumeExhaustThreshold
    (very few contracts at bar extreme = last buyer/seller =
exhaustion)
```

V3_volume (Sufficient total volume):

```
+1 IF BarVolume >= MinTurnVolume (default: 0.5 x AvgBarVolume)
```

V4_price_action (Reversal candle):

```
Bullish Turn: +1 IF BarClose > BarOpen + (BarRange x 0.5)
Bearish Turn: +1 IF BarClose < BarOpen - (BarRange x 0.5)
(bar closed in the reversal direction by >50% of range)
```

V5_delta (Delta confirmation):

```
Bullish Turn: +1 IF BarDelta > 0 (positive delta supports
bounce)
Bearish Turn: +1 IF BarDelta < 0 (negative delta supports drop)
```

V6_swing (At swing high/low):

```
+1 IF BarHigh >= SwingHigh_N - 2_ticks (bearish turn at swing
high)
    OR BarLow <= SwingLow_N + 2_ticks (bullish turn at swing
low)
```

Turn Signal Emission

IF TurnScore >= 3 AND TurnScore <= 4: TurnStrength = MODERATE

IF TurnScore >= 5: TurnStrength = STRONG

TurnDirection:

BULLISH_TURN = BarLow is at extreme AND V4 = bullish reversal candle

BEARISH_TURN = BarHigh is at extreme AND V4 = bearish reversal candle

17. Orderflows Ratio — Top Heavy / Bottom Heavy

9.1 Formula

For detecting BOTTOM HEAVY (Big Buyer at Low):

```

LowPriceBidVol  = bid_vol[BarLow_price]
LowPriceAskVol  = ask_vol[BarLow_price]
BarLow_total    = LowPriceBidVol + LowPriceAskVol

BottomHeavyRatio = LowPriceBidVol / max(LowPriceAskVol, 1)

BottomHeavy = TRUE
  IF BottomHeavyRatio >= RatioThreshold (default 3.0 = 300%)
  AND BarLow_total    >= MinRatioVolume (default 100 contracts)
  AND BarLow is within N_ticks of SessionLow (default N = 5)

```

For detecting TOP HEAVY (Big Seller at High):

```

TopHeavyRatio = HighPriceAskVol / max(HighPriceBidVol, 1)

TopHeavy = TRUE
  IF TopHeavyRatio    >= RatioThreshold
  AND BarHigh_total   >= MinRatioVolume
  AND BarHigh is within N_ticks of SessionHigh

```

★ One signal fires per bar maximum. BottomHeavy and TopHeavy are mutually exclusive per bar — a bar cannot be both simultaneously. If both conditions are met (unusual), the stronger ratio takes priority.

18. Single Prints — Last Buyer / Last Seller

10.1 Formula

```

LastBuyer_at_High = TRUE
  IF total_vol[BarHigh_price] <= SinglePrintThreshold (default 2
contracts)
  AND ask_vol[BarHigh_price] > 0 (someone bought at the high)
  AND BarHigh >= PriorBarHigh (new high or equal high)

LastSeller_at_Low = TRUE
  IF total_vol[BarLow_price] <= SinglePrintThreshold
  AND bid_vol[BarLow_price] > 0 (someone sold at the low)
  AND BarLow <= PriorBarLow (new low or equal low)

ExhaustionScore = 1.0 - (total_vol[extreme_price] /
AvgVol_at_extremes)
  where AvgVol_at_extremes = EMA(total_vol[BarHigh_price], 10)

UnfinishedAuction = TRUE
  IF ask_vol[BarHigh_price] > 0 (ask still showing at high →
unfinished)
  OR bid_vol[BarLow_price] > 0 (bid still showing at low →
unfinished)

```

19. Delta Surge

11.1 Formula

AvgDelta_N = EMA(**ABS**(**BarDelta**), **N**) (N-bar EMA of absolute delta, default N=14)

SurgeMagnitude = **ABS**(**BarDelta**) / **max**(**AvgDelta_N**, 1)

BullishSurge = **SurgeMagnitude** >= **SurgeThreshold** AND **BarDelta** > 0

BearishSurge = **SurgeMagnitude** >= **SurgeThreshold** AND **BarDelta** < 0

SurgeThreshold = user-configurable (default 3.0 = 300% of average delta)

BarsBuilding = count of consecutive bars with **ABS**(**BarDelta**) >= **AvgDelta_N**

(how many bars in a row have above-average delta in the same direction)

Example: AvgDelta_14=200, BarDelta=+750 → SurgeMagnitude = 750/200 = 3.75 → BullishSurge triggered (3.75 >= 3.0)

20. Market Sweep Detector

12.1 Formula

A market sweep occurs when a large aggressive order clears through multiple consecutive price levels in rapid succession within a single bar.

$\text{SweepVolume} = \text{SUM}(\text{dominant_vol}[P])$ for P in sweep range
 where $\text{dominant_vol}[P] = \text{ask_vol}[P]$ for bullish sweep, $\text{bid_vol}[P]$ for bearish

$\text{SweepPriceLevels} = \text{count of consecutive price levels cleared in one direction}$

$\text{SweepSpeed} = \text{SweepPriceLevels} / \text{SweepTimespan_ms}$ (levels per millisecond)

$\text{Sweep_detected} = \text{TRUE}$
 IF $\text{SweepVolume} \geq \text{SweepVolumeThreshold}$ (default 500 contracts)
 AND $\text{SweepPriceLevels} \geq \text{SweepLevelThreshold}$ (default 5 price levels)
 AND $\text{SweepSpeed} \geq \text{SweepSpeedThreshold}$ (default 2 levels/ms)
 AND all levels cleared in same direction (no opposing volume > 10% of sweep)

21. Trapped Traders Detection

13.1 Formula

```
TrappedSellers = TRUE
  IF SellingImbalance detected at price P
  AND P is within 2 ticks of BarLow
  AND BarClose > BarOpen + (BarRange x 0.5)    (bar closed in upper
half)
  (Sellers who sold the low are now trapped – price closed strongly
above them)

TrappedBuyers = TRUE
  IF BuyingImbalance detected at price P
  AND P is within 2 ticks of BarHigh
  AND BarClose < BarOpen - (BarRange x 0.5)    (bar closed in lower
half)
  (Buyers who bought the high are now trapped – price closed strongly
below them)

TrapStrength = ImbalanceRatio_at_level x (BarRange / AvgBarRange)
```

22. POC Slingshot

14.1 Formula

```
POC_Slingshot = TRUE (BULLISH)
  IF PriorBar: BarClose within N_ticks of SessionPOC (price near POC)
    AND CurrentBar:
      BarClose > SessionPOC + SlingDistanceTicks x tick_size
      (price broke away from POC by at least SlingDistance ticks)
    AND BarDelta > SlingDeltaThreshold x AvgBarDelta
      (strong delta supporting the break – default 1.5x average)
    AND BarVolume > SlingVolumeThreshold x AvgBarVolume
      (above-average volume confirming the break – default 1.2x average)

SlingDistanceTicks = user-configurable (default 4 ticks for ES)

POC_Slingshot_Bearish = symmetric definition – break BELOW POC
```


23. Complete Formula Reference Summary

Indicator	Core Formula	Key Variables	Default Threshold
Bar Delta	ask_vol_sum - bid_vol_sum per bar	ask_vol[P], bid_vol[P] at each price level	N/A — raw value
Delta Percentage	BarDelta / TotalBarVolume × 100	BarDelta, TotalBarVolume	N/A — raw %
Max/Min Delta	max/min of RunningDelta_t within bar	RunningDelta_t per tick	N/A — tracking extremes
Cumulative Delta	CVD_n = CVD_{n-1} + BarDelta_n	BarDelta sequence	Resets at session open
VWAP	CumPV / CumVol from anchor	CumPV = $\sum(P \times V)$, CumVol = $\sum V$	Session open as anchor
VWAP Std Dev	$\text{SQRT}(\text{CumPV}^2 / \text{CumVol} - \text{VWAP}^2)$	CumPV ² , CumVol, VWAP	k=1,2 for bands
Lee-Ready	Quote Rule → Tick Rule	P_t, M_t, Bid_t, Ask_t, P_{t-1}	P_t vs M_t first
Buying Imbalance	ask_vol[P] ≥ ratio × bid_vol[P-1tick]	ask_vol[P], bid_vol[P-1]	ratio = 3.0 (300%)
Selling Imbalance	bid_vol[P] ≥ ratio × ask_vol[P+1tick]	bid_vol[P], ask_vol[P+1]	ratio = 3.0 (300%)
Stacked Imbalance	≥ N consecutive imbalances same direction	ImbalanceChain, N	N ≥ 3 consecutive
Absorption	total_vol ≥ thresh AND delta ≤ ratio × total_vol	total_vol[P], delta[P]	500 contracts, 10%
POC	argmax(profile[price])	volume histogram	Highest volume level
Value Area	Expand ±2 levels from POC until ≥ 70% volume	profile[], target_vol	70% (configurable)
HVN	profile[P] ≥ mult × mean_vol	profile[P], mean_vol	mult = 1.5
LVN	profile[P] ≤ mult × mean_vol	profile[P], mean_vol	mult = 0.5
COT Price	argmax(total_vol[P]) within bar	ask_vol[P]+bid_vol[P]	Per-bar calculation
Pulse Score	Weighted sum of 7 sub-scores (0-100)	7 variables, weights sum to 1.0	Score ≥ 70 = signal
Turns Score	Count of 6 binary variable alignments (0-6)	6 binary conditions	Score ≥ 3 = signal
Orderflows Ratio	ask_vol[High] / bid_vol[High] for TopHeavy	extreme price volumes	ratio ≥ 3.0

Single Print	$\text{total_vol}[\text{extreme}] \leq \text{threshold}$	volume at bar extreme	≤ 2 contracts
Delta Surge	$\text{ABS}(\text{BarDelta}) / \text{AvgDelta_N} \geq \text{threshold}$	BarDelta, EMA(BarDelta ,14)	≥ 3.0 (300%)
Market Sweep	Sweep volume + levels + speed thresholds	vol, level count, speed	500 contracts, 5 levels
Trapped Traders	Imbalance at extreme + opposite close direction	Imbalance price, BarClose	Within 2 ticks of extreme
POC Slingshot	Break from POC + delta + volume confirmation	POC, BarDelta, BarVolume	4 ticks, 1.5× avg delta
Delta Divergence	Price new high/low + CVD opposite direction	Price_n, CVD_n vs prior bars	Configurable lookback k

END OF SPECIFICATION — v4.0

v4.0 adds Section 9–23: Complete Indicator Calculation Formulas for all 26 indicators

Sections 1–8: Core Spec (orderflows.com, references, Volume Profile, API, timestamps, bar config) | Sections 9–23: All indicator formulas